

HYBRID TRANSPARENCY: An Open-Source ZK-SNARK Model for Public Accountability

by

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Bachelor of Science in Computer Science with Specialization in Software Engineering

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APPROVAL AND ACCEPTANCE SHEET

The thesis entitled “Hybrid Transparency: An Open-Source ZK-SNARK Model for Public Accountability” prepared and submitted by:

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(OPTIONAL)

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TABLE OF CONTENTS

THESIS TITLE	i
APPROVAL AND ACCEPTANCE SHEET	iii
ACKNOWLEDGMENT	iv
TABLE OF CONTENTS	v
LIST OF TABLES	vii
LIST OF FIGURES	viii
Chapter 1	9
INTRODUCTION	9
1.1. Background of the Study	11
1.2. Statement of the Problem	14
1.3. Objectives of the Study	15
1.4. Theoretical Framework	16
1.5. Conceptual Framework	26
1.6. Significance of the Study	30
1.7. Scope and Delimitations	32
1.8. Definition of Terms.....	33
Chapter 2	38
REVIEW OF RELATED LITERATURES	40
2.1. Foreign Literatures	40
2.2. Local Literatures	45
2.3. Synthesis	46
REVIEW OF RELATED STUDIES	48
2.4. Foreign Studies	48
2.5. Local Studies	53
2.6. Synthesis	54
Chapter 3	56

METHODOLOGY	58
3.1. Research Type	56
3.2. System Design	57
3.3. System Architecture	63
3.4. Hardware and Software Requirements	64
3.5. Methods in Developing Software	66
3.5.1 Scrum/Agile Methodology	66
3.5.2 Algorithm	67
3.6. Methods in Evaluating the System	71
3.7. ISO Based Software Evaluation (ISO/IEC 25010)	76
BIBLIOGRAPHY	79
APPENDICES	

LIST OF TABLES

Table 1	64
Table 2	64
Table 3	65
Table 4	65
Table 5	65
Table 6	72
Table 7	74

LIST OF FIGURES

Figure 1.4.1	16
Figure 1.4.2	18
Figure 1.4.3	19
Figure 1.4.4	20
Figure 1.4.5	21
Figure 1.4.6	24
Figure 1.5.1	26
Figure 1.5.2	28
Figure 3.2.1	58
Figure 3.2.2	60
Figure 3.2.3	61
Figure 3.3.3	62
Figure 3.3.4	63
Figure 3.5.1.....	66
Figure 3.5.2.....	66
Figure 3.5.2.1.....	68
Figure 3.5.2.2.....	69
Figure 3.5.2.3.....	70
Figure 3.5.2.4.....	71

Chapter 1

INTRODUCTION

Blockchain adaption has been increasingly rampant around the world. With EU's law passed, all 27 member states must be able to accomplish a cross-platform digital identity system called eIDAS 2.0 (Electronic Identification, Authentication Service) by 2026. This new regulation implements the EU Digital Identity (EUDI) wallet that must be open source, setting a global standard for digital trust.

Recently, the Philippine government has also launched major blockchain initiatives. This includes the "eGov" Dapp by DICT, which holds user credentials, "Project Marissa" by the DBM for budget transparency (Mislos, 2023), and "Project Sierra" by DOST-ASTI for SSI (Self-Sovereign Identity) (Valcorza, 2024). This presents a landscape of contrasting approaches: the EU's mandated open-source model versus a local trend of innovative, but closed-source, government-led projects.

However, this issue is not limited to the Philippines, it is a global concern. Currently, there is no widely accepted international standard for evaluating blockchain frameworks, particularly in governance. Many government initiatives simply digitize existing manual, paper-based processes, turning them into digital formats without embracing the core principles of open-source transparency. As a result, modernization becomes superficial, and the public still cannot see what happens behind the scenes.

Countries like China (Kumar, 2025) and the UAE (Морозова, 2024) are building systems that prioritize state control, effectively hiding the inner workings from the public. Even Estonia, uses software that isn't open for everyone to check (Mäeots, 2022). The danger of this 'closed-door' approach is clear in places like Nigeria, where the eNaira became a surveillance risk because it lacked privacy features (Basu, 2021). Basically, without open standards, these systems demand blind trust instead of offering proof.

The research uses the Self-Sovereign Identity (SSI) Trust Model from the Verifiable Credentials Data Model v2.0 to solve the security issues that centralized identity management systems face. The SSI system establishes different operational functions through its design, which requires all verification requests to be handled by the issuing authority. The study follows the model requirement that the issuer does not need to know anything about the verifier, according to (Sporny et al. 2024).

The proposed system allows the citizen (holder) to possess complete ownership of their credential after the institution provides it. The citizen performs a proof demonstration to a third party (verifier), which establishes a direct link between the holder and verifier. The system uses cryptographic decoupling to prevent the issuing institution from tracking citizen identity usage because it transforms the system from a centralized trust model to a cryptographic trust model, which protects user data through protocol-based security measures.

To achieve this, this project attempts to use architectures from the Ethereum Foundation itself and apply them to a government framework that allows for the best possible transparency and user security. By utilizing privacy methods used previously in Digital Finance, user identity is kept secure while letting the auditors track system changes, effectively balancing institutional security and public accountability. We will implement an architecture that can be applied in a multitude of public sectors.

1.1. Background of the Study

The "closed source" approach seen in the Philippine “eGov” and other project initiatives, while innovative, creates a critical "black box" problem. This forces 110 million citizens and developers to trust the operator, not the technology, an approach that runs counter to the ethos of decentralization. According to recent research by (Bappy et al. 2025), the complexity of decentralized systems creates a “digital gap,” which means that non-technical users are more vulnerable to online manipulation and scams due to their lack of understanding. This creates a dangerous "Digital Literacy Gap," where lawmakers and the public are prone to being misled, unable to distinguish a truly verifiable system from one that simply uses "blockchain" as a buzzword.

Blockchain is about decentralization and security, where many high-profile platforms have failed due to “black boxes”. Instead of providing an open, mathematical verifiability system, these platforms forced users to blindly trust the technology by hiding centralized control and internal mechanisms from public view. For example, the collapse

of QuadrigaCX exchange (OSC Staff 2020) and the Multichain protocol (Copeland 2023) occurred because their closed-source architectures granted sole control of private keys to a single administrator, creating a fatal single point of failure that could not be audited by the public. Another example is platforms like Celsius Network and FTX (Reiff 2025) that failed because their black box structures allowed executives to secretly commingle billions in customer deposits and conceal highly risky lending practices until the platforms went bankrupt.

In one extreme case of failed blockchain technologies due to black box, is the OneCoin scandal. The lack of a transparent cryptographic anchor allowed operators to completely fabricate a cryptocurrency on a centralized SQL database. These failures prove that when blockchain systems rely on hidden and centralized logic rather than transparent and cryptographic proofs, it inevitably turns the users' own trust against them and becomes highly vulnerable to systemic collapse and fraud.

In 2020, World Economic Forum (WEF) established a partnership with inspector general of Columbia to create a test pilot for a public procurement transparency. In an aim to mitigate corruption, vendors participated in an open bidding process on the blockchain, where transaction data was made visible to other participants and auditors to prevent fraud. But during this experiment, the WEF failed to hide the private details of the vendors, thus exposing them to the public, because the payment of the transaction fees on the public blockchain can be traced back to the real identities of the vendors.

The experiment in Colombia may be a failure, but in recent world events, blockchains have been used by various countries. For example, in China, the government has implemented several systems, such as Blockchain-based Service Network (BSN) and Tian Ping. According to (WIPO 2022), the Tian Ping system allows the Beijing Internet Court to verify digital evidence by comparing its on-chain cryptographic hash with the off-chain original file. This strategy allows the Chinese judiciary to settle millions of cases with “mathematically proven” evidence integrity. However, it is critical to note that this auditability is distinct from public decentralized networks. China’s government utilizes a “permissioned” model where audit rights are restricted to authorized government nodes rather than the public.

In Estonia, the government made a solution through blockchain that ensures the integrity of national data networks against cyberattacks and internal corruption. According to (e-Estonia n.d.), the government created the Keyless Signature Infrastructure (KSI) technology that allows the government to guarantee data authenticity without compromising privacy and security. This model also allows only the state or government to audit, which is another example of a closed-source system.

While countries like Estonia and China have their governments involved in blockchain, the Philippines has its own problem with the rapid digitalization of public services, where a large part of the gap in 'Public Trust' architecture is yet to be fulfilled. Current digitalized public services function as centralized 'black boxes' where the public has no assurance of data integrity, and high-risk data breach by the government or any high-power individual. In addressing this situation, according to research by (Bappy et al. 2025),

argue that separating public and private implementations is critical for institutional adoption, as private layers are necessary to protect sensitive confidentiality while public layers handle transparency.

Additionally, blockchain frameworks nowadays rely on the ‘Triangle of Trust’ model. (Sporny et al., 2025) Defined this model by mentioning a core component of Self-Sovereign Identity (SSI) standards, where a single entity acts as both custodian and validator, decentralizing trust into three distinct roles. The Issuer, which is the authority that creates the credential; the Holder, which is the individual who retains power over their data in a secure wallet; and lastly, the Verifier, which is a third party capable of validating claims independently. This type of decentralized structure allows verifiers to mathematically confirm the validity of a credential against a public anchor without requiring direct access to the issuer’s private database.

1.2. Statement of the Problem

Traditional Blockchain lacks abstractions and compliance, as seen in the World Economic Forum (WEF) transparency paradox. Current adoption resorted to highly controlled and centralized private blockchains. Which abstracts “Gas Fees”, making it attractive for governance. Given blockchains transparent nature, its auditability has not been used to its potential. Furthermore, we see from (Bappy et. al) that blockchain remains enigmatic to most. This lack of literacy heightens the dangers imposed by “black box” and closed-source initiatives. While there are no reports of incidents in

government “black box” implementations yet. The trust with blockchain system can be eroded, destroying the impression that blockchain promises.

- How can the system enforce compliance while maintaining user anonymity during interactions?
- How can the system establish a trust model that separates the issuer from the verifier to protect against institutional surveillance and unauthorized data modification?
- What system architecture enables transparency and public verifiability without exposing sensitive data?
- How can the system serve as a functional blueprint to help students, teachers, developers, and policymakers understand the contrast between "closed-source" and "open-source" governance systems?

1.3. Objectives of the Study

General Objectives:

The objective of the study is to address the "black box" problem in current centralized applications by implementing Ethereum frameworks and the W3C Verifiable Credentials v2.0 Standard. The study adopts the W3C triangle of trust that separates public accountability from private user data, ensuring that government records can be mathematically audited by the public without compromising the sensitive personal information of citizens. This project selects drivers licensing as its

simulation due to its familiarity and commonly used identity framework in traditional governing systems. Integrating this infrastructure, it envisions a futuristic governmental deployment aligned with modern standards and Data Privacy laws.

Specific Objectives:

- **To utilize the Semaphore v4 Protocol and ZK-SNARKs as a compliance layer.**
Simulating a Self-Sovereign Identity (SSI) wallet, user generates proof as part of compliance and Proof of Inclusion. Merkle trees is used to store user commitments that enable users to prove their license by providing Merkle Paths without being revealed. This one-time commitment allows users to generate proofs for on chain and off chain purposes.
- **To integrate the W3C Verifiable Credential Data Model v2.0.** It must be architecturally separated from the Issuer from the Verifier. this prevents institutional surveillance and ensures that the issuing authority cannot track or monitor the usage of citizen credentials. Allowing user to generate proofs without calling the institution or Database for validating credentials.
- **To implement a Smart Contract, serving as an immutable verification log.**
Admin interacts by inserting and revoking users in the Merkle Tree. The National Chain is assumed “open network”; thus, anyone can become a “watcher node” or view transactions in a block explorer. Allowing independent auditors to spot inconsistencies in the registry instantly. Furthermore, Validium architecture is implemented by anchoring the state root of the Merkle Trees onto a public chain. This prevents erasures by forking blocks on the National Chain.

- **To simulate a complete, privacy-preserving LTO licensing compliance workflow in a National Chain (Private Chain).** It will serve as a technical blueprint for future developers, students, and policymakers, demonstrating how decentralized systems can aid the "digital gap" in understanding blockchain technology.

1.4. Theoretical Framework

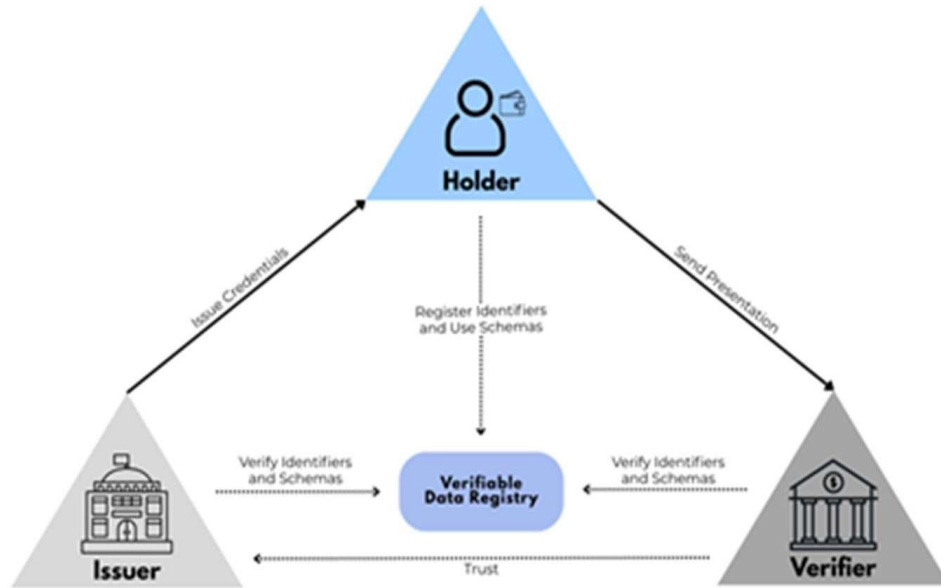


Figure 1.4.1 Verifiable Credentials Data Model v2.0: Triangle of Trust

The theoretical framework for Figure 1.4.1 illustrates the Triangle of Trust in the context of Self Sovereign Identity (SSI), which is foundational to this research. This model standardize by the World Wide Web Consortium (W3C), decentralizes traditional trust mechanisms by defining three distinct roles: the Issuer, who creates and issues verifiable credentials; the Holder, and individual who maintains complete control and ownership over their digital identity and credentials; and the Verifier, a third party capable of independently validating the authenticity of a credential without direct interaction with the Issuer’s private database. This SSI-based Triangle of Trust directly benefits this research by providing a strong framework for enhancing public accountability and user security within digital systems. By empowering Holders with cryptographic control over their data and enabling Verifiers to confirm claims through public anchors, the model mitigates the risks associated with centralized identity management. This cryptographic ensures privacy by preventing the tracking of identity usage, by means that transforming a centralized trust model into a

more secure and transparent cryptographic one, which is crucial for the development of a hybrid transparency model that balances institutional security with public error.

Building upon the foundational roles of Issuer, Holder, and Verifier, Figure 1.4.1 introduces the Verifiable Data Registry as a pivotal component that underpins the entire trust ecosystem. This registry, often implemented as a decentralized ledger or blockchain, serves as a public and immutable source for registering and resolving Decentralized Identifiers (DIDs) and credential schemas. Its function is critical: the Holder utilized this registry to register their DIDs and reference the schemas for their credentials, ensuring that these digital identities are publicly discoverable and verifiable. Concurrently, both the Issuer and the Verifier interact with the Verifiable Data Registry to verify the authenticity of DIDs, and the integrity of the schemas used for issuing and presenting credentials. This shared transparent infrastructure eliminates the need for direct, bilateral trust between the Issuer and Verifier, as both parties can cryptographically confirm the validity of identifiers and credential formats against a common tamper-proof source. The Verifiable Data Registry acts as the cryptographic anchor, enabling the decentralized verification process and reinforcing the trust within the SSI Triangle by providing a robust, auditable mechanism for identity and credential management.

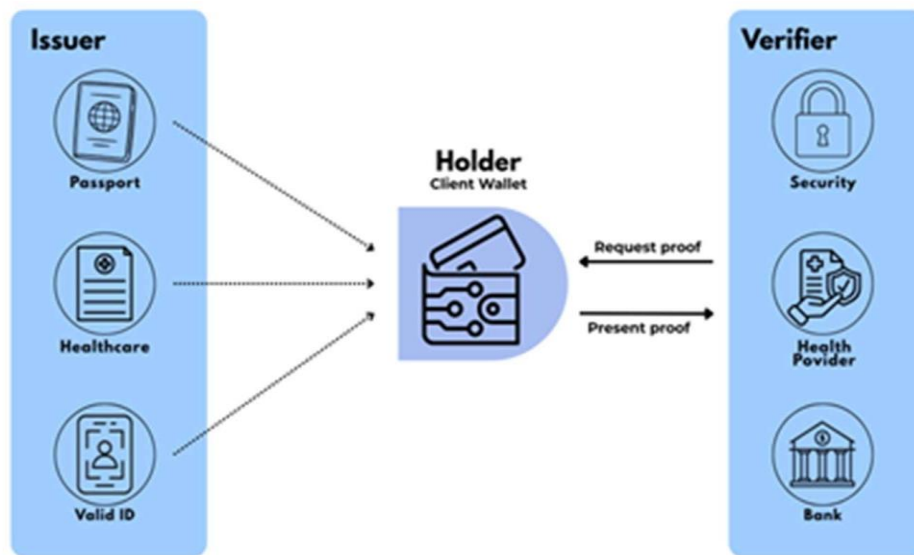


Figure 1.4.2. Self-Sovereign Identity

This Figure 1.4.2 diagram shows the operational mechanics of Self-Sovereign Identity (SSI) within the Triangle of Trust framework. It illustrates the three core roles: the Issuer, represented by entities like passport authorities, healthcare providers, or government agencies, which issue verifiable credentials to the Holder. The Holder, equipped with a “Client Wallet,” maintains complete control and ownership over these digital credentials. When interacting with a Verifier, such as security checkpoints, health providers, or banks, the Holder receives a “Request proof” and subsequently provides a “Present proof” from their wallet. This interaction signifies a key benefit of SSI: the Verifier can independently validate the authenticity of the presented credential without needing to directly query the original Issuer, thereby decentralizing trust and enhancing privacy. This practical representation reinforces the theoretical underpinnings of SSI, demonstrating how it empowers individuals with precise management over their personal data, streamlines verification processes across diverse domains, and ultimately contributes to a more secure

and accountable digital identity ecosystem, aligning directly with the research’s objectives improving accountability and user security.

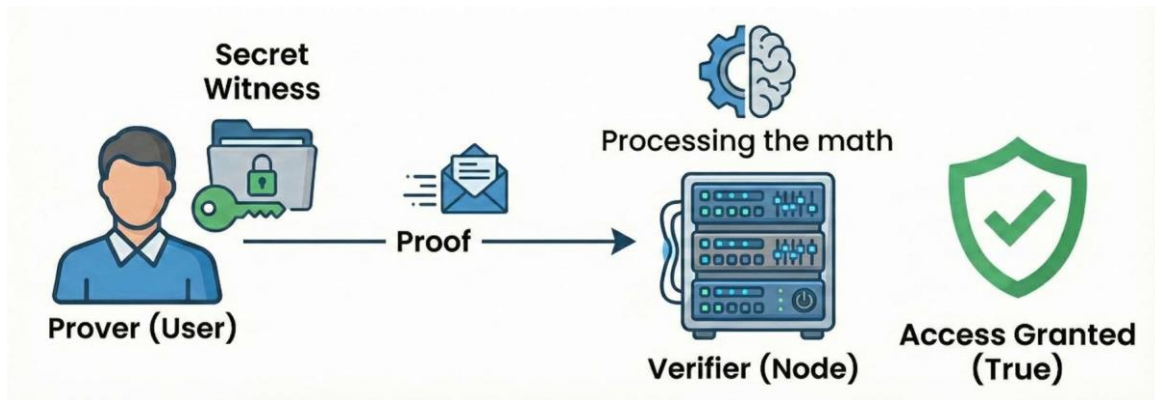


Figure 1.4.3. Zero-Knowledge Proof

This Figure 1.4.3 is theoretically supported by Goldwasser, Micali, & Rackoff’s work in the year (1989). Figure 1.4.3 shows how Zero-Knowledge Proofs or ZKP, as a method for proving a statement’s truthfulness without exposing privacy. We apply the mathematical concept to a functional system using ZK-SNARKs (Zero Knowledge Succinct Noninteractive Argument of Knowledge), which in turn uses the ‘Noir’ domain-specific language.

The verification process is strictly followed under the regulation of three mathematical concepts necessary in trustless systems. Completeness is maintained to ensure that all citizens who have a right to the claim, such as individuals who possess a legitimate credential, always pass through the verification process successfully. Soundness is simultaneously used to ensure that no fraud attempt occurs to convince the verifier to claim otherwise.

Lastly, Zero-Knowledge Proof ensures that the verification process reveals nothing beyond the validity of the statement itself. There is a certain conformity on the part of the

system regarding ensuring the truth about a citizen's eligibility while maintaining secrecy about the information and never disclosing it. In accordance with these principles, it keeps the system fully compatible with the requirements laid down under the Data Privacy Act of 2012, which allows citizens to validate his/her identification and eligibility without exposing his/her personal data.

Formally, the theoretical foundation of this protocol is defined by the universal zero-knowledge mathematical statement: $\exists w: C(x, w) = 1$ This denotes that there exists a private witness (w) such that a verification circuit (C) evaluates to true (1) for a given set of public inputs (x), without revealing the witness itself. This foundational equation ensures that the system can verify an individual's eligibility purely through mathematical logic rather than data exposure.

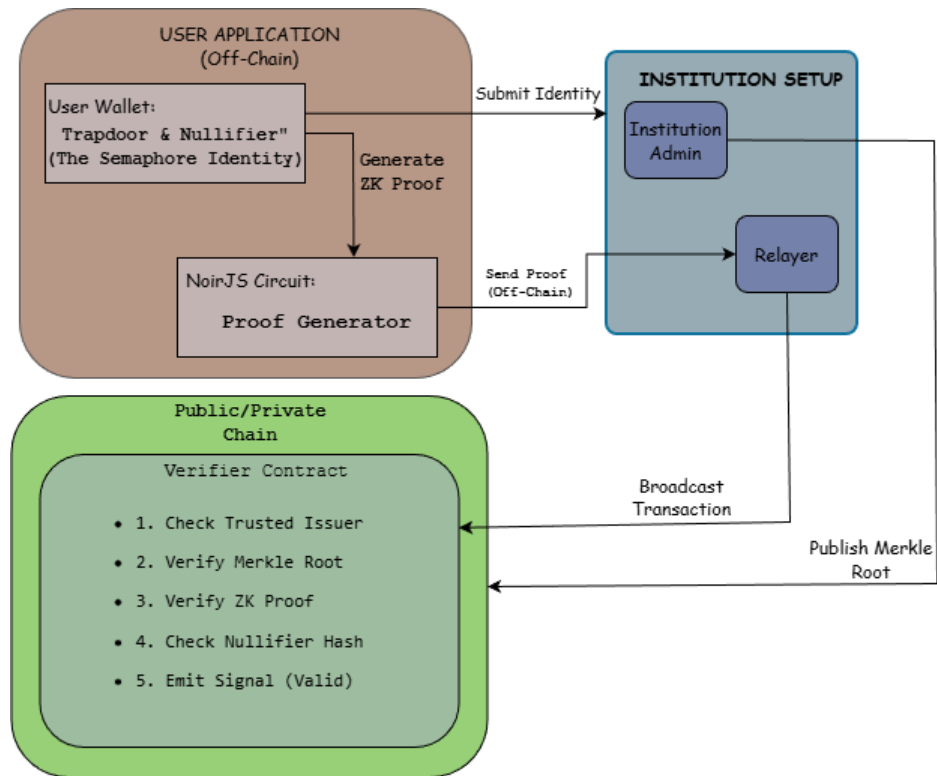


Figure 1.4.4. Semaphore: Zero-Knowledge Signaling on Ethereum

This Figure 1.4.4 show that it is also grounded in the theoretical concepts of Semaphore, Roberts et. al from the year (2019). They designed the Semaphore Protocol to offer a system for "Anonymous Membership Verification," as it enabled users to verify their group identity without revealing any personal identity. Here in this research, the "Signaling" concept is used in a functional system by relying on the v4 architecture of the protocol, that is, maintained by the Privacy & Scaling Explorations PSE team of the Ethereum Foundation.

This is done by following a verification process that is strictly governed using cryptographic "Nullifiers" needed in a trustless environment. This also maintains accountability that is significant to ensure that every valid action, including the action by an official in approving a license, is written into the ledger. The principle of Uniqueness is

also employed at this stage in preventing double signaling by any official through the processing of a transaction twice.

Most importantly, the rule of privacy applies in this case too. So, there is some compliance of the system about the authenticity of the government process and the matter of keeping it a secret who the individuals are. Accordingly, following such guidelines will enable us to show that our "Hybrid Transparency" framework works and that accountability and identification can be separated from each other mathematically.

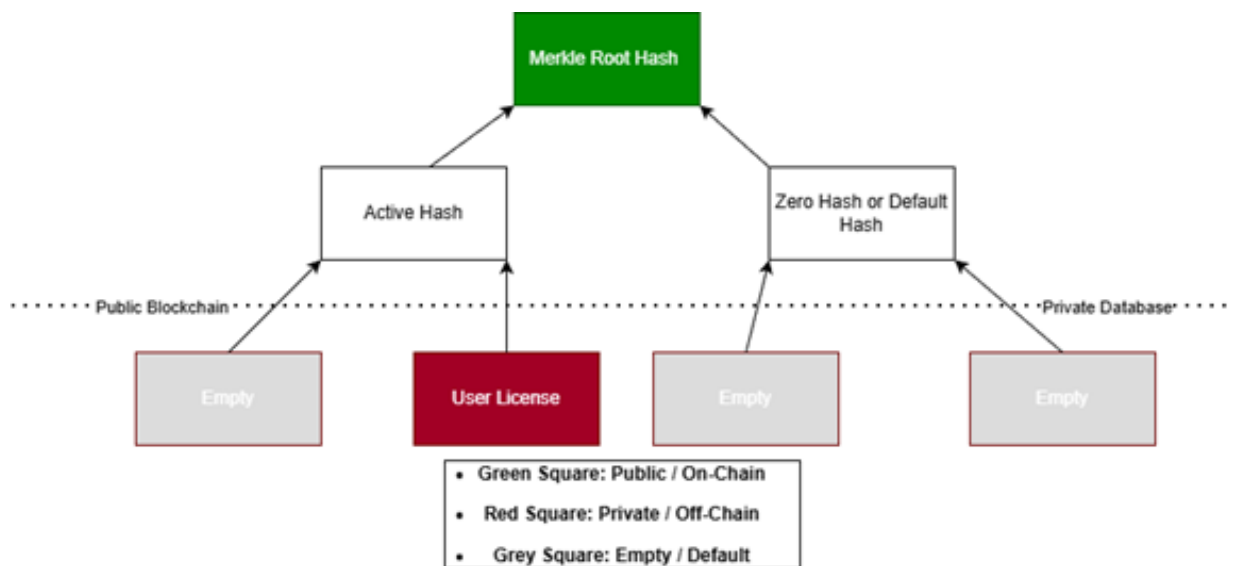


Figure 1.4.5. Sparse Merkle Trees in Blockchains: Building Efficient Proofs

Figure 1.4.5 shows Sparse Merkle Trees, where it attributes our blockchain system to its ability to manage large-scale data sets efficiently, such as those that will apply as license registrars. According to Maya Navarre (2025), these cryptographic trees allow for the commitment of massive datasets into a single root hash. In this framework, the Merkle root serves as the public "Anchor" on the blockchain, while individual citizen data remains private. This structure enables one to perform an efficient proof of inclusion, which means that the system will be able to verify whether a particular credential is present within the

authorized dataset with no need to search through any sort of centralized "black box" database.

To further utilize the verification efficiency, the system will specifically use a Sparse Merkle Tree. According to Dahlberg, Pulls, and Peeters (2016), the Sparse Merkle Tree structure allows the system to simulate a large-scale 'perfect' tree of theoretically intractable size efficiently. It can recognize that the majority of leaves are empty and can be represented by a recursive default hash value. This means that it can enable the generation of Proof of Non-Inclusion that allows the verifier to mathematically confirm that a revoked or invalid credential does not exist in the database. Additionally, a different and more recent study by Xu, Li, and Erkin (2023) mentioned that if a user's license expires or an identity collision is detected, the system updates that specific data or leaf node to a revoked value. This update changes the global Markle Root by ensuring that any attempt by the 'revoked' user to generate another valid proof will ultimately fail against the new public anchor, making the blockchain more secure.

This type of architecture is critical for the system's "Hybrid Transparency" because it allows for the execution of an efficient Proof of Inclusion. This mechanism enables any verifier to mathematically confirm that a specific credential exists within the authorized government registry by simply tracing its hash path to the root, eliminating the need to query or trust a centralized, opaque "black box" database. Most importantly, this ensures that the integrity of the data is tamper-evident, meaning any unauthorized alteration to a record in the private database would change the root hash, instantly alerting the public network.

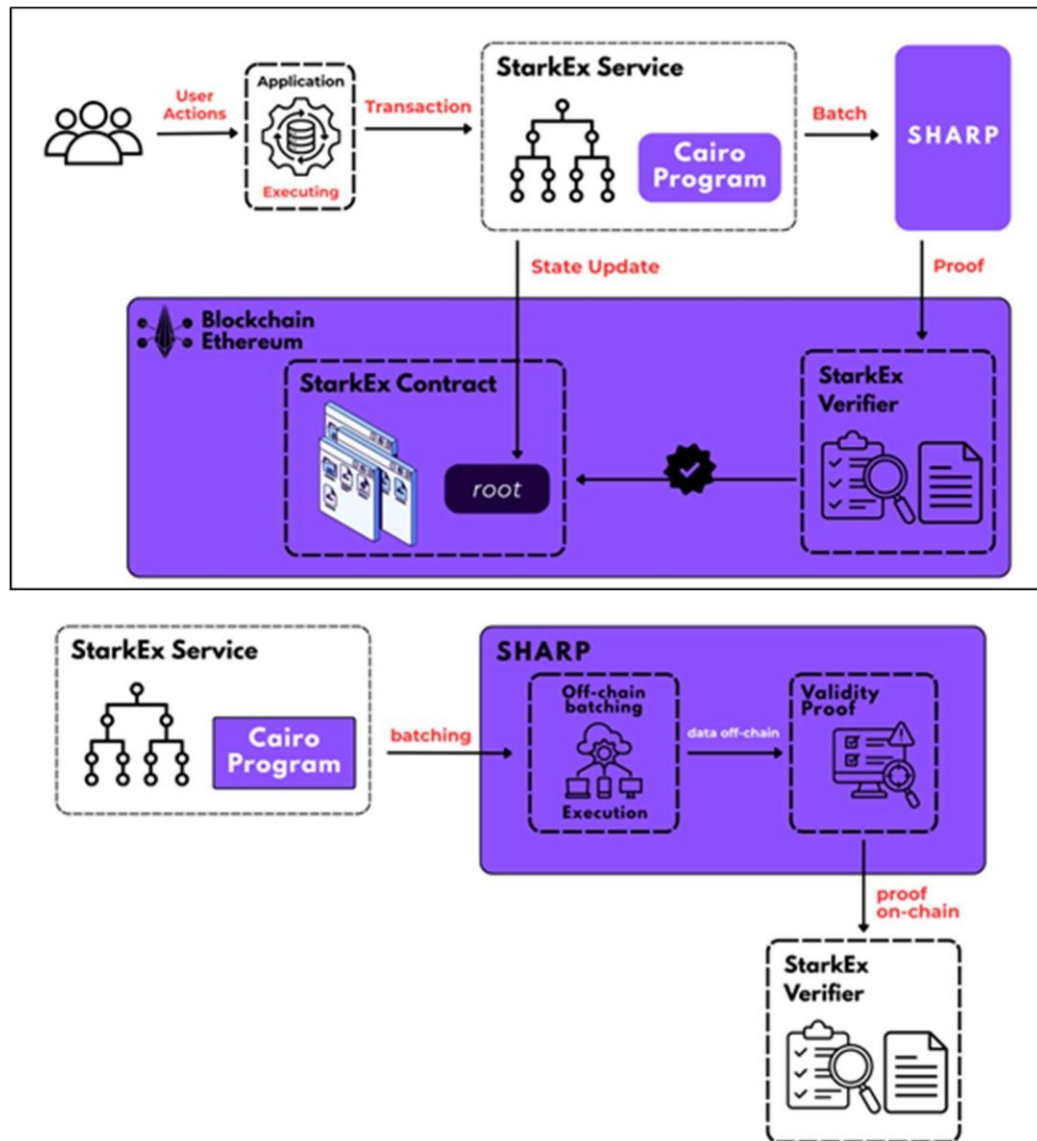


Figure 1.4.6. Validium

This study is theoretically supported by StarkWare's (2020) work. The Figure 1.4.6 shows that the authors proposed the Validium scaling architecture that aims to improve blockchain performance by reducing the amount of data stored directly on the Ethereum network while still maintaining strong security guarantees. In this approach, transaction processing and data storage are handled off-chain, which significantly lowers gas fees compared to executing every transaction on Ethereum. Validium only stores a

cryptographic summary of the system's state instead of recording all transaction data on chain. This is also represented by a Merkle tree root known as the state root, which securely reflects all account balances and system data within the network.

State updates are handled by an operator who processes transactions off-chain and generates a new Merkle root that represents the updated system state. This new state root is submitted to an Ethereum smart contract along with validity proof, and once it is confirmed to be correct, the new state is accepted as the official system state. In this way, Validium ensures that all state changes are mathematically verified while using Ethereum as the final source of trust. This design enables high transaction throughput and low fees without sacrificing security, making it well-suited for large-scale decentralized applications.

1.5. Conceptual Framework

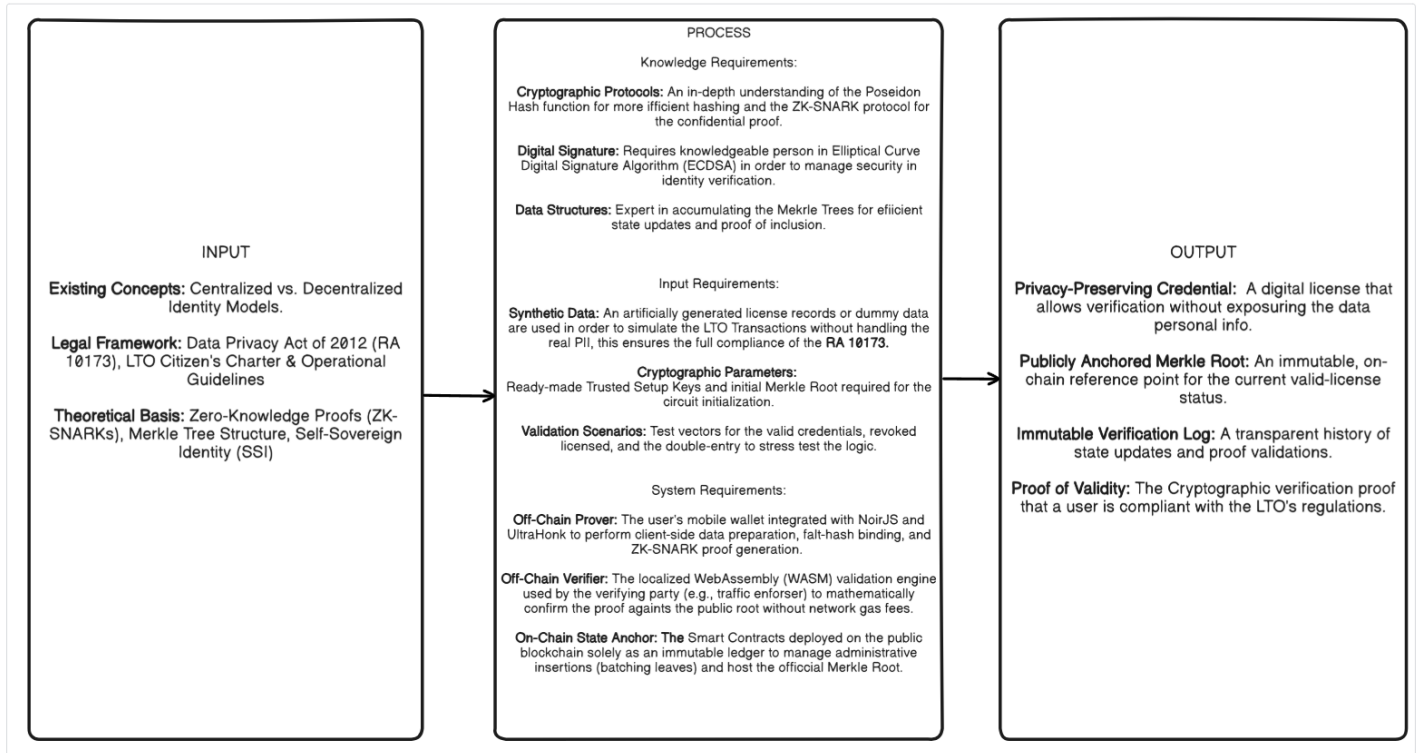


Figure 1.5.1 Conceptual Framework

Figure 1.5.1 illustrates the Input-Process-Output (IPO) process diagram of the proposed privacy-preserving licensing architecture. The input begins by establishing the foundational requirements for the study. This includes the existing LTO Citizen's Charter and Operational Guidelines, which dictate the standard licensing workflow, and the Data Privacy Act of 2012 (RA 10173), which enforces strict legal constraints on handling personal data. Additionally, the theoretical basis is established using Zero-Knowledge

Proofs (ZK-SNARKs) and Self-Sovereign Identity (SSI) models to replace traditional, vulnerable centralized databases.

The Process serves as the operational engine and is categorized into three core requirements. First, the knowledge requirements define the necessary technical expertise, specifically utilizing the Poseidon Hash Function for efficient cryptographic data binding, and the Elliptic Curve Digital Signature Algorithm (ECDSA) to deterministically derive the user's high-entropy Semaphore v4 secret. Second, the input requirements mandate the strict use of synthetic data to simulate LTO verification without violating privacy laws, alongside the validation scenarios necessary for stress-testing valid credentials and replay-attack (double-entry) rejections. Third, the system requirements define the decoupled architecture: an Off-Chain Prover (NoirJS) handles client-side proof generation, an Off-Chain Verifier allows authorities to validate credentials locally via WebAssembly, and an On-Chain State Anchor utilizes the public blockchain strictly to host the immutable Merkle Root and manage administrative leaf insertions.

Lastly, the output stage produces the privacy-preserving credential following standard off-chain KYC (Know Your Customer) procedures. The issuing administrator binds the user's cryptographic secret with their credential parameters (e.g., License Number, Name) via a flat hash to generate an Identity Commitment (Leaf), which is subsequently inserted into the smart contract's Sparse Merkle Tree. When prompted, the system produces a verifiable payload containing a ZK-SNARK proof and the specific, selectively disclosed public data.

This architecture successfully balances state governance with individual privacy, illuminating the traditional "black box" of agency operations. Public accountability does not arise from the mass disclosure of individual user data, but from the ability of any external observer to independently verify that all institutional state transitions are correct, complete, and immutable through publicly anchored Merkle roots and uncompromising mathematical verification logic.

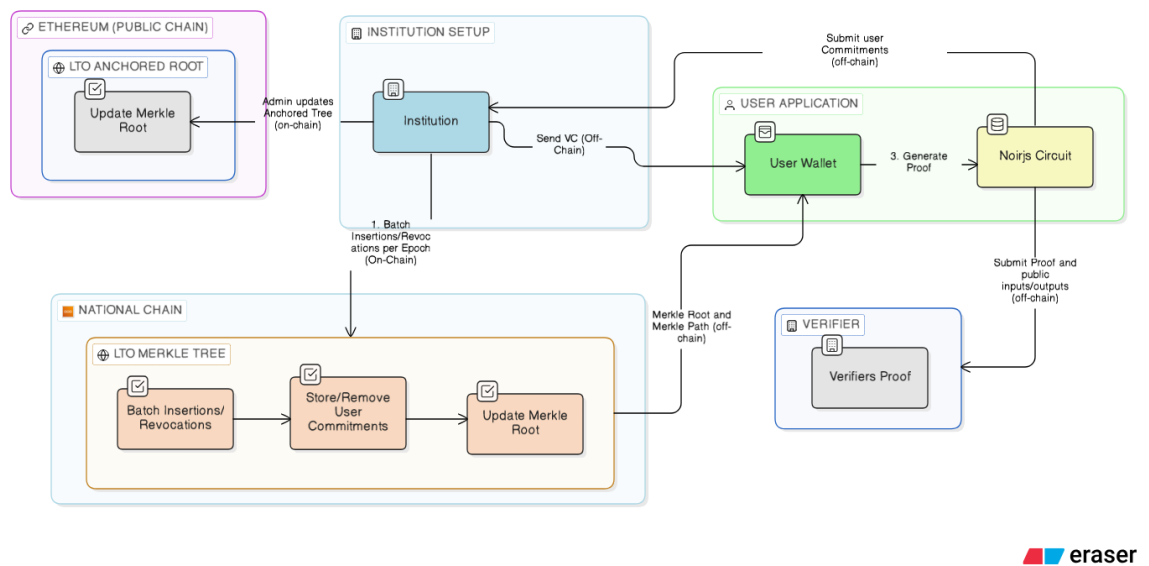


Figure 1.5.2 Hybrid Semaphore v4 and Validium

As shown in Figure 1.5.2, the system establishes a dual-chain cryptographic boundary designed to meet enterprise and WEF procurement standards for public accountability. This hybrid public/private architecture utilizes the modernized Semaphore v4 zero-knowledge signaling protocol, enabling users to prove their eligibility without exposing personally identifiable information (PII) to either chain.

The user initiates the process during issuance through client-side data preparation. The mobile wallet derives a high-entropy cryptographic **Secret** from a digital signature. The institution administrator computes a flat-hash Identity Commitment (Leaf) binding this Secret to the user's credential data. This commitment is then inserted directly into the **National Chain**—a permissioned, private blockchain acting as the secure operational base. The National Chain securely stores the Sparse Merkle Tree containing all valid users, protecting the hashed commitments from public exposure.

During the Proof Generation phase, the user's mobile device utilizes a NoirJS circuit to construct a Zero-Knowledge Proof. This client-side execution mathematically verifies the user's inclusion in the National Chain's Merkle Tree while dynamically generating a scoped Nullifier to prevent replay attacks. The user transmits this proof payload to the verifying authority (e.g., a traffic enforcer). The verifier subsequently queries the National Chain to validate the proof against the active Merkle Tree and record the Nullifier.

To ensure systemic transparency without compromising data privacy, the architecture implements a Validium scaling solution. While the National Chain serves as the Trusted Availability Layer—processing high-throughput credential verifications and managing the Merkle Tree internally—it periodically anchors its State Root to the **Public Chain** (e.g., Ethereum Sepolia). This public Ethereum contract acts purely as an immutable reference point. It ensures that while specific citizen data remains off-chain and localized to the private consortium, the overall integrity and state transitions of the government's registry can be independently audited by the public.

1.5.3 Formal Zero-Knowledge Statement of the System

At a theoretical level, the proposed Hybrid Transparency system is defined by a specific zero-knowledge proof statement that allows public verification of institutional compliance without the disclosure of sensitive citizen data. Formally, the client-side NoirJS circuit mathematically proves that the user owns a valid credential embedded within the government's registry by evaluating the following exact constraints:

The Mathematical Constraints:

1. **Leaf Reconstruction:** $L = \text{Poseidon}(s, d, N)$
2. **Nullifier Derivation:** $\text{Nullifier} = \text{Poseidon}(s, S)$
3. **Merkle Inclusion:** $R = \text{MerklePath}(L, p)$
4. **State Verification:** $R_{\text{computed}} == R_{\text{public}}$

Data Classification:

To evaluate these constraints, the system strictly divides the credential parameters into hidden and public categories:

- **Private Witnesses (Data being hidden):** The user's cryptographic secret scalar (s), the sensitive credential data (d) such as the exact license number, and the array of Merkle sibling hashes (p) required to traverse the tree.

- **Public Inputs (Data accessible to the public):** The anchored state Merkle Root (R_{public}), the selectively disclosed credential data (N) such as the user's name, and the Scope (S), which acts as a unique deterministic challenge message for the current verification session.
- **Public Outputs:** The generated *Nullifier* and the ZK-SNARK proof payload.

Justification for Public Accountability via Data Minimization: While the systemic transparency of this architecture is derived from the public state root of the smart contract, the act of actively hiding specific user data (Private Witnesses) is what ensures true public accountability. In traditional centralized architectures, auditing a system requires the mass exposure of sensitive Personally Identifiable Information (PII), creating severe security vulnerabilities. By isolating parameters such as the user's cryptographic secret and exact credential string as Private Witnesses, the system enforces the principle of data minimization. It allows the issuing authority to publicly prove the unforgeable integrity of its registry to the blockchain, while simultaneously protecting the citizen from unwarranted surveillance and data harvesting. Therefore, hiding this specific data benefits public accountability by decoupling system verification from citizen exposure—achieving an auditable government registry without compromising individual privacy.

1.6. Significance of the Study

The significance of this study lies in demonstrating how privacy-preserving technologies can be applied to real-world government processes without sacrificing transparency or accountability. By simulating the Land Transportation Office (LTO)

licensing workflow using zero-knowledge proofs, Merkle verification, and modular opensource components, the study presents a practical technical model for auditable yet privacy compliant public services. This work bridges theoretical cryptographic concepts and applied system design, showing how existing legal and institutional frameworks can support verifiable digital governance. These findings provide useful guidance for real world implementation. The study contributes to several key groups:

- General Public – This study addresses the "Blockchain Trust Paradox" (Keaney & Berthon, 2025), which shows that users refuse to trust completely decentralized systems because these systems lack both "experienced trust" and methods to identify responsible parties. The system establishes a Hybrid Architecture to solve the paradox by giving the public an actual institution (LTO) which they can use to demand accountability for service delivery, while the system maintains its cryptographic transparency. Citizens can depend on the institution for assistance, but they also possess the mathematical methods to verify their records against unauthorized alterations, which guarantees governmental responsibility to the public.
- Teachers – This study provides teachers with an LTO licensing simulation as a teaching tool, featuring code blocks and verification steps that clarify ZK-SNARKs for classroom. Utilizing diagrams outline flows like use cases for license checks, enabling interactive lessons. Teachers can demonstrate Merkle tree proofs in real time; it turns complex privacy topics into practical experiments.
- Students – Students engage hands-on with the LTO model, adjusting ZK proofs and tracing credential verification from start to finish. They may build skills using open

code designed for CS/IT learners; it serves them as a complete blueprint to begin exploring the blockchain technology.

- **Developers** – Developers receive open-source code from the LTO simulation that integrates Project Sierra's selective sharing and Marissa's budget transparency, handling off-chain data smoothly. This toolkit makes privacy-compliant coding straightforward and scalable; reusing modules for any public workflow, slashing the build time.
- **Policymakers** – The LTO simulation gives policymakers hard evidence via ZKSNARKs aligned with the Data Privacy Act by enabling audits on license approvals; envisioning rules where government proves fairness without exposing the data of the citizens. It breaks down how selective proofs maintain trust in public eyes.
- **Public Sectors** – While this zero-knowledge architecture utilizes the Land Transportation Office (LTO) as its primary simulation model, the framework is designed for general application across any public sector agency that handles sensitive citizen data. Furthermore, cryptographic proofs guarantee non-repudiation, ensuring no party can deny completed transactions and directly eliminating "black box" concerns.
- **Future Researchers** – Future researchers can extend the LTO framework to identity systems or decentralized audits using its tested code and baselines. This advances privacy-focused governance research. The documentation highlights areas like state proofs for further study; it must support scaling for national applications.

In essence, this simplifies ZK-SNARKs' application to public systems, equipping future developers with a roadmap for privacy-preserving, accountable infrastructure that aligns with national digital transformation goals.

1.7. Scope and Delimitations

This study focuses on the design and development of a technical prototype for decentralized credential issuance and verification system operating under an assumed national private blockchain infrastructure, where citizens already possess government-issued e-wallets. The scope begins after identity proofing, presuming that all users and issuing institutions such as LTO and hospitals have already undergone secure, real-world verification processes.

The system is developed as an architectural and functional proof of concept that simulates legal and regulatory compliance through business logic, without engaging in formal legal analysis. It emphasizes the core zero-knowledge proof workflow, including the use of ZK-SNARKs via the Semaphore protocol. ZK-STARKs provide advantages for increased system capacity and elimination of required trusted initialization procedures, but their implementation demands more processing power and data storage than this prototype can handle. The research restricts its scope to ZK-SNARKs because they demonstrate efficient proof size performance and their existing tools support Proof of Concept (PoC) development.

The architecture includes the anchoring of Merkle roots to a public blockchain while utilizing local storage, dummy wallet addresses, and placeholder data to avoid

handling real identities or sensitive records. Consequently, the project is explicitly delimited from operational cybersecurity, real DID/SSI wallet integration, legal enforcement mechanisms, quantum-resistant cryptography, and high-concurrency scalability testing (e.g., simulating millions of simultaneous users).

Furthermore, this system lacks auditable software, as it's simulated under hardhat Environment. Unlike in real world deployment, etherscan for example, is used to read Ethereum blocks. Hardhat Environment already provides commands in the system, allowing developers to read each block. These elements are acknowledged as essential for real-world deployment but are excluded to align with the project's limited timeframe, ethical constraints, and focus on delivering an architectural blueprint rather than a production-ready system.

1.8. Definition of Terms

The blockchain and cryptography fields do not yet have universally standardized terminology. Concepts such as unlinkability, anonymity, zero-knowledge, membership proofs, and decentralization may vary in meaning across different research communities and implementations. For clarity and consistency, this study adopts the definitions explicitly stated in this section. All terms are used in the context defined herein, regardless of variations in industry or academic usage.

Anchor – The process of storing the Merkle Root (the unique "fingerprint" of the entire group) onto the public blockchain. The system creates an unchangeable checkpoint which enables off-chain data validation without requiring complete member data to be stored on the blockchain.

Blackbox – abstracted inner workings of a system design to hide implementations from the public.

Blockchain – is a digital ledger that everyone can see, but the records of it cannot be changed; it simply provides people with a source of truth.

Bootstrapping – A self-reliant process used for initializing a secure system or network.

Circuit - The mathematical "blueprint" or logic that defines the rules of a zero-knowledge proof. The system uses specific conditions (such as "Is this user in the Merkle Tree?") The system needs all generated proof to meet these constraints.

Cryptographic – refers to the mathematics used to protect the integrity of information from unauthorized access.

Dapp – Decentralized Applications. The software application functions through blockchain networks instead of using a single central server. The system depends on smart contracts which execute its functions automatically while maintaining transparent operations and preventing unauthorized modifications and single entity control.

Ethereum – It is one of the most renowned decentralized and open-source blockchain platforms that offer the capability of smart contracts and Dapps (Decentralized Application) development.

Merkle Path – The Merkle Root of a specific leaf node requires multiple sibling hashes to be reconstructed. The system uses cryptographic proof to demonstrate the existence of specific data, such as a user's ID, which exists in the database without needing complete database access.

Merkle Proof – A ‘path’ of cryptographic hashes that proves a specific piece of data exists within a Merkle Tree. The system can verify user membership through the Merkle Root which enables them to maintain user information security by not requiring access to their complete user database.

National Chain – Similar to Layer 2 (L2) solution or “Side chain”, it’s a conceptual government blockchain that acts as an execution layer.

Nullifier Hash – A unique cryptographic value which emerges from a transaction function as an anonymous digital receipt. The system uses this capability to identify and reject duplicate operations which include double voting. The system achieves this function while maintaining complete privacy for all user information.

Off-chain – Refers to the processing of transactions, data, that are recorded or stored in a centralized network or end device.

On-chain – Refers to transactions or pieces of data processed, recorded, or stored on the main blockchain network.

Semaphore v4 – An architecture for zero knowledge proofs algorithms interacting with the blockchain for proving users' inclusion without being revealed.

Validium – Scalability solution which stores the transaction data off-chain and the validation of the Validium proofs is carried out on the chain.

Zero-knowledge Proof – This refers to a cryptographic process through which one can prove the truth of a claim without revealing the information to which it refers.

ZK-SNARK – Zero Knowledge Succinct Non-Interactive Argument of Knowledge. A cryptographic protocol that enables users to verify their possession of specific data through their actual valid identification without showing their identification.

ZK-STARK – Zero Knowledge Scalable Transparent Argument of Knowledge. The Zero-Knowledge Proof (ZKP) system provides two main benefits by using "Transparent" to eliminate the requirement of a trusted setup and "Scalable" to generate proofs more quickly during large-scale computations. The system creates bigger proof outputs which increase its storage costs on blockchain networks when compared to SNARKs.

Chapter 2

REVIEW OF RELATED LITERATURES

This chapter introduces and describes both local and foreign literature, as well as local and foreign studies. The following sections of this chapter will show all research results through

detailed study of research contributions and investigative methods and research findings which will help develop a complete understanding of all scholarly work which exists in both local and international research fields.

2.1 Foreign Literatures

2.1.1 Exploring Blockchain Technology for Government Transparency:

Blockchain-Based Public Procurement to Reduce Corruption.

This study by the World Economic Forum (2020) explores how blockchain technology can be used to improve transparency in government and reduce corruption. In addition, this paper is the first to take a multidimensional approach to blockchain experimentation specifically for anti-corruption. The authors aim to contribute to the global understanding of how public-sector transparency can be improved through technology, using their report as a case study to highlight the real-world issues, limitations, and policy considerations involved.

The report advocates for a "hybrid" blockchain network which employs both a permissioned and a permissionless layer as the most feasible solution. According to their statement, the permissioned configuration refers to an "invite-only" or private environment, similar to the off-chain (LTO database) section in our study, where sensitive data is kept controlled. Whereas the permissionless configuration refers to fully open public access, which aligns with our on-chain verification layer. The authors conclude that this hybrid system is the most suitable solution because it offers an ideal balance between transparency, procedural integrity, and security.

However, the idea was eventually unable to be fully implemented due to specific limitations in the technology available at the time. The report shows a critical vulnerability: on a public blockchain, paying transaction fees can reveal a vendor's identity. To hide this, the team used "hidden IDs" funded by a central pool. Unfortunately, this funding pool was vulnerable to "draining attacks", where outsiders could steal the funds meant for legitimate fees. The study admits that "because of the PoC's constrained software development timeline, this vulnerability exists in the end solution," leading to a standstill in deployment.

This failure clearly demonstrates the importance of our project. The WEF study's goals are identical to ours, but they failed because the required cryptographic tools were not yet ready. Unlike their solution, which depended on a vulnerable "central pool" to hide personal information, our research utilizes Zero-Knowledge Proofs (ZKP). This allows us to give them the privacy they desired while reducing the security issues they faced, ultimately closing the gap that prevented their solution from succeeding.

2.1.2 Can Blockchain Make the National Budget Tamper-Proof? (this is a local bro)

Domingo (2025) discusses how the recent flood defense scandal in the Philippines exposed long-standing problems of corruption, where funds marked for critical infrastructure vanished into ghost projects because acquisition and payout processes operated with little transparency. This article argues that embedding the national budget on a blockchain would allow every peso spent to be recorded on a immutable, timestamped, and auditable ledger, making it far harder for officials to conceal misallocations or fabricate projects because all transactions could be checked directly by auditors and other regulators,

as well as ordinary citizens, if allowed (Domingo, 2025). This study directly illustrates the black box problem highlighted in the present study, demonstrating that opacity in financial systems is not an abstract concern but a concrete driver of real-world losses in public funds. On the other hand, an open-source ZK-SNARKs-based architecture, such as the proposed research, would enable stakeholders to cryptographically verify the correctness of budget-related transactions without exposing sensitive information, shifting the system from one that relies on trust in centralized disclosures to one grounded in mathematically provable integrity.

2.1.3 QuadrigaCX: A review by staff of the Ontario Securities Commission.

According to OSC Staff (2020), the QuadrigaCX scandal serves as a definitive case study illustrating the inherent fragility of “closed” architectures. The report details how millions in user funds vanished overnight because the exchange operated as a 'black box,' with the founder holding sole control over the private keys while internal operations remained completely opaque to depositors. This incident directly supports the thesis's argument regarding the critical vulnerability of asking the public to trust a system they cannot verify. The collapse demonstrates that 'closed source' platforms create a catastrophic single point of failure, verifying the research premise that reliance on the unverified honesty of a central administrator is a fundamental security flaw rather than a viable governance model. OSC Staff (2020) illustrates that when a system relies entirely on the reputation of a central administrator rather than on verifiable proofs, it creates a single point of failure that is vulnerable to both internal fraud and mismanagement. Consequently, OSC Staff (2020) underscores that the fundamental flaw lies not just in poor management, but in the

architectural reliance on 'blind trust,' proving that opacity prevents stakeholders from auditing the true integrity of the system until the damage is irreversible.

2.1.4 Celsius Was Lying

In this article of Bloomberg “Celsius Was Lying” (July 14, 2023), Matt Levine, a columnist, characterizes Celsius Network as having fundamentally misrepresented its business model and financial operations by promising high, sustainable returns to customers while concealing the true riskiness and unsustainability of its lending practices, culminating in regulatory lawsuits and criminal charges against its executives for fraud after the platform froze withdrawals and collapsed into bankruptcy. Regulators alleged that Celsius made numerous false and misleading statements about the security of customer assets, the nature of its lending, and its ability to generate returns, effectively drawing customers in under false pretenses. It illustrates a black box problem in financial systems such as the internal mechanisms, risk exposure, and operational realities were opaque to users and external observers, such that stakeholders could not meaningfully assess how customer deposits were being used or how promises were being fulfilled, leaving the platform’s inner workings inscrutable until failure, which aligns with how opaque “black box” structures conceal critical information and amplify systematic risk.

2.1.5 Manhattan U.S. Attorney Announces Charges Against Leaders Of “OneCoin”, A Multibillion-Dollar Pyramid Scheme Involving the Sale of A Fraudulent Cryptocurrency.

According to (U.S. Attorney's Office 2019) during 2016, Ruja Ignatova launched "OneCoin" with the main purpose of becoming the "Bitcoin Killer," a financial revolution that would change cryptocurrencies. However, this vision was later revealed to be a catastrophic "Black Box" tragedy when technical investigations confirmed that OneCoin lacked an actual blockchain, operating instead on a centralized SQL database where administrators could manually simulate coin balances and transactions. This lack of public verifiability allowed Ignatova to orchestrate a massive Ponzi scheme that defrauded investors of an estimated \$4 billion before she vanished in October 2017. The OneCoin collapse serves as a reminder of how "blind trust" can effectively destroy the financial stability of people, and illustrates how digital infrastructures without a public, cryptographic anchor can weaponize user trust to maintain an illusion of value while masking systemic administrative fraud.

2.1.6 Multichain ceases operations as CEO's sister is detained in China while holding \$220 million of funds.

This literature illustrates the risk of relying on a single team for security, according to Copeland (2023) regarding the sudden collapse of the Multichain protocol. This literature tells how the platform of the Multichain shut down in July 2023 after its CEO, Zhaojun, and his sister were obtained by the police in China. This event exposed a massive red flag in the system because the operation, including the \$220 million in user funds, was actually being controlled by just one family. When Zhaojun and his sister got arrested, the rest of the development team couldn't access and repair the system or even retrieve the money.

Copeland (2023) points out that before the shutdown, money started mysteriously moving out of the accounts, which looked like a hack but couldn't be proven because the internal workings were hidden from the public. This disaster highlights exactly why our study is necessary. The Multichain case proves that when a system's safety depends on trusting a few people behind closed doors, it is prone to sudden collapse. In contrast, this research proposes a mathematical solution using Zero-Knowledge Proofs. Instead of trusting an administrator who could be arrested or compromised, our system uses cryptographic Merkle Roots to enforce the rules. This ensures that the verification process remains open and honest, regardless of what happens to the people running it.

2.1.7 FTX Crypto Exchange Collapse: Causes, Consequences, and Lessons.

In 2019, Sam Bankman-Fried founded FTX with the goal of building a leading global cryptocurrency exchange that appeared secure, liquid, and professionally managed. However, in November 2022, this image collapsed as investigations revealed that FTX had secretly funneled and commingled billions of dollars in customer deposits into its affiliated trading firm Alameda Research using internal control failures and misleading balance sheets to conceal the true state of its finances. Reports from the subsequent bankruptcy proceedings indicated that at least 8 billion dollars in client funds had been misappropriated, contributing to one of the most significant financial frauds in recent history and triggering a broader crisis of confidence across the crypto industry. The FTX controversy underscores how opaque centrally controlled platforms can exploit user trust when there is no on-chain transparency or independent verifiability of reserves, reinforcing calls for stronger oversight and more transparent, auditable infrastructure in digital asset markets (Reiff, 2025).

2.1.8 Ice Phishing Scams.

The Ice Phishing is described as a highly targeted form of blockchain-based scam in which attackers' trick Web3 users into manually signing transactions that grant spending permissions on their tokens rather than stealing passwords or private keys outright. By disguising malicious smart contract approvals as legitimate interactions with decentralized exchanges, metaverse platforms, or customer support links, these scams exploit the fact that most users cannot interpret low-level transaction data and end up “blind signing” complex permissions they do not fully understand. This phenomenon illustrates the wider danger of opaque systems: if ordinary users are already losing funds because they cannot verify what they authorize on chain, the potential impact becomes even more severe when similar opacity surrounds critical domains such as public budgets or sensitive institutional data (Kevin Moore, 2022). The case of Ice Phishing supports the study that blind trust in hidden logic is a structural vulnerability rather than a desirable future, and it motivates the need for mechanisms like open-source ZK-SNARK-based verification, where cryptographic proofs can validate the correctness of a transaction before approval and shift protection from user intuition to mathematically verifiable guarantees.

2.2 Local Literatures

2.2.1 Bitkwela launches Philippines’ first blockchain-enabled microlearning platform.

Blockchain technology entered the Philippine educational sector that shifts the architecture from a traditional “trust-based” systems to a “verification-based” system. An example is the Bitskwela, according to (TechNode Global, 2025), which recently launched the Philippines' first blockchain-enabled microlearning platform to address the workforce skills gap. This initiative utilizes distributed ledger technology to issue immutable “proofs of learning,” allowing employers to instantly validate a candidate's credentials without relying on manual background checks or third-party intermediaries. Instead of using the static way, it will now be easily forged paper certificates with cryptographically secured digital assets. The platform demonstrates the practical application of a “verification-based” architecture in a local setting, proving that Philippine institutions are ready to adopt decentralized standards for data integrity.

2.2.2 Blockchain and Government Efficiency.

The systemic inefficiencies within Philippine government agencies are critically examined in the policy note by the Philippine Institute for Development Studies (PIDS, 2021). The report emphasizes the Land Transportation Office (LTO) as an example of outdated data management, criticizing the agency’s persistent reliance on people to manually input applicants’ information into computers. PIDS argues that this "menial" manual process is the primary reason for the constantly long driver's license lines and creates significant opportunities for "fixers" and human error to corrupt the system.

Furthermore, PIDS (2021) warns that the current government practice of maintaining "unprotected and non-transparent silos" creates a dangerous paradox: these closed databases are highly vulnerable to data breaches yet remain completely opaque to

the public, preventing any independent audit. This finding highlights the specific gap that the current study fills: the need for "Hybrid Transparency."

Our proposed system directly resolves the conflict observed by PIDS. While the PIDS report states that agencies lock data away to protect it, our research implements a Zero-Knowledge Proof (ZKP) architecture that offers a secure "hybrid" solution. To prevent breaches, sensitive raw data such as the applicant's name and address are stored off-chain in the secure LTO database. Simultaneously, a cryptographic "proof of validity" is published on-chain to ensure transparency. This effectively eliminates the "menial" manual verification criticized by PIDS; instead of calling the LTO or accessing the private database to check a license, they can simply verify the on-chain proof, achieving the efficiency of a public ledger without exposing the private data inside.

Synthesis

These literatures have supported how blockchain can integrate into our government but also mentions the risk of it.

To start off, (TechNode Global, 2025) notes that their blockchain, "Bitskwela" has somehow made it easier for employers to effectively validate a candidate's credentials without the need for manual checking. (Broby, 2025) also mentioned how blockchain transparency can lessen the incidents of having "ghost projects". These literatures support how blockchain can benefit the country once the government fully understands how to utilize it.

However, if one can't fully grasp how blockchain works, its importance, and its dangers, it would lead to catastrophic failures in creating centralized 'black box' systems, and history proves it. One of the failures is best exemplified by the QuadrigaCX scandal, where (OSC Staff, 2020) reported that the platform's opaque structure allowed \$250 million in user funds to vanish because the CEO held sole control over the private keys. Without a transparent ledger to prove solvency, stakeholders were left helpless when the founder died. This pattern of 'blind trust' failure is further verified by (Levine, 2023), who details how the Celsius Network collapsed after executives misled users about the riskiness of their lending practices, which led to deception that was only possible because the internal mechanisms were hidden from public view.

Furthermore, the risks of blockchain extend beyond mismanagement to outright fabrication. (U.S. Attorney's Office, 2019) where it revealed that the OneCoin made by Ruja Ignatova defrauded investors of billions by operating on a centralized SQL database while pretending to use a blockchain, which directs to effectively simulating value without any cryptographic anchor. Similarly, the collapse of the Multichain protocol (Copeland, 2023) demonstrates that even systems claiming to be secure are vulnerable if they rely on a single team or family holding the keys. These instances collectively validate the thesis argument: that without the cryptographic verification tools proposed in this study, such as ZK-SNARKs, stakeholders are forced into a vulnerable position of "blind signing" transactions they cannot audit, leaving them exposed to both internal fraud and systemic collapse.

REVIEW OF RELATED STUDIES

2.3 Foreign Studies

2.3.1 The Impact of Security and Privacy Perception on Cryptocurrency by Users: A text Mining Study.

The researchers Juma'h, Alnsour, and Kartal (2025) examined 64000 user evaluations to determine why cryptocurrency software receives lower customer satisfaction ratings when compared to standard banking applications. The research showed that negative assessments stemmed primarily from users' security and privacy worries rather than any software faults or design deficiencies. Public adoption of blockchain technology requires that all applications must show protective measures which safeguard user information because perceived risk represents the greatest obstacle which users encounter when they try to adopt new technology.

The Hybrid Transparency model which this thesis presents receives direct validation from this finding. The research conducted by Juma'h et al. (2025) discovered that users stop using their services because of two factors, which include system blackouts and security threats. The system uses Hybrid Semaphore v4 to transform "blind trust" into "cryptographic proof" according to its design. The system provides citizens with mathematical verification methods which they can use to prove their data protection because it connects to the public blockchain, which Juma'h et al. identified as the essential failure point in decentralized applications.

2.3.2 Centralized Trust in Decentralized Systems: Unveiling Hidden Contradictions in Blockchain and Cryptocurrency.

The limitations of current decentralized interfaces are highlighted in the study by Bappy et al. (2025). The authors argue that blockchain technology still requires significant improvement in usability, pointing out that the lack of user-friendly designs often forces users to trade their privacy for the convenience of a centralized "safety net." The research specifically points out that users often stick to centralized trust simply for accountability, rationalizing that "if something goes wrong, at least I know who to blame." Therefore, the paper proposes a hybrid framework that combines the security of decentralized networks with the accountability structures of centralized platforms.

This proposal aligns directly with the objectives of our study. Like the hybrid model suggested by Bappy et al., this research uses the centralized authority of the Land Transportation Office (LTO) with the decentralized security of zero-knowledge proofs. As a result, the system achieves the balance of fairness, accountability, and transparency and ensures that users have a clear "safety net" for their license validity without sacrificing their data privacy to a "Black Box" database.

2.3.3 User Mental Models of Cryptocurrency Systems – A Grounded Theory Approach.

The disconnect between user expectations and technical reality is explored in the study by Mai et al. (2020) regarding user mental models of cryptocurrency. These researchers highlight that while decentralized systems aim to remove central control, they often expose users to dangerous security and privacy risks, including fraud, deanonymization, and loss of funds due to fundamental flaws in how users understand the technology. Specifically, the study revealed a critical inconsistency: many users mistakenly

believed that blockchain data is "encrypted" or "oblivious", which leads them to risky behaviors under the misconception that their privacy is already safe.

Mai et al. (2020) conclude that to prevent these breaches, system architects must improve the usability of management tools to better align with these mental models. This recommendation is the basis of our study. This research closes the gap identified by Mai et al. by implementing Zero-Knowledge Proofs (ZKPs). Unlike the traditional blockchains, where privacy is often misinterpreted, our proposed system validates the user's "mental model" of privacy. By using ZKPs to cryptographically hide the raw data (converting it into a hash/proof) before it touches the public chain, the system ensures that the user's belief that their data is hidden and secure is no longer a misconception but a mathematical fact.

2.3.4 Leveraging zero knowledge proofs for blockchain-based identity sharing: A survey of advancements, challenges and opportunities.

Diro et al. (2024) presents an in-depth investigation of the value of cryptographic privacy in decentralized systems by examining the inherent conflict within blockchain technology: while the public ledger provides transparency and stability, it also violates user privacy by revealing transaction histories. The study concludes that Zero-Knowledge Proofs (ZKPs) are the most promising solution to this paradox, as they allow data validity to be verified without revealing sensitive data.

Diro et al. (2024) really encourage the development of new frameworks, saying that "further research addressing this gap could offer innovative approaches for leveraging zero-knowledge proofs to enhance the security and privacy of identity management systems."

This gap is the main goal of our research. Responding to the authors' need for innovative solutions, this research applies ZKP technology directly to the Land Transportation Office

(LTO) framework. By utilizing Merkle Proofs and ZK-SNARKs, the proposed system allows citizens to prove the validity of their license to the public chain without exposing their raw personal data (such as their age). This basically creates the "privacy-preserving identity management system" that the literature identifies as the necessary next step for blockchain technology adoption.

2.3.5 Adoption of e-Government Services in the Northern Part of Cyprus: The Role of Blockchain Technology Awareness.

A study published in Sage Open, (Çoban et al. 2024) investigates the critical factors influencing e-government adoption. Identifying that trust in the internet and performance expectancy are the primary factors for citizen engagement. While their research focuses on the behavioral aspects of digital governance, they highlight a significant barrier, where even when infrastructure exists, citizens are still hesitant to use centralized systems, as they perceive a lack of security or transparency. Additionally, (Çoban et al. 2024) argue that to effectively increase adoption rates, government institutions must prioritize the deployment of systems that guarantee transaction security and strictly protect user privacy against potential security breaches.

2.3.6 The Dark (Patterns) Side of UX Design.

Grey et al. (2018) highlight the risks of relying on user interfaces for privacy consent in their study on "Dark Patterns". The researchers show how centralized systems often use manipulative design strategies such as "sneaking" and "forced action" to manipulate users into giving away more personal data than they intended. This study shows

that in a traditional Web 2.0 setting, "user consent" is often an illusion, as the interface can be weaponized to prioritize data extraction over user action.

This issue demonstrates why the recommended cryptographic framework is important. While Grey et al. showed how visual interfaces can be weaponized to fool users, this study prevents that risk by changing the security model from the "screen" to the "protocol" using Zero-Knowledge Proofs (ZKPs). By implementing privacy regulations directly into the system's architecture, the LTO framework ensures that data privacy is mathematically enforced rather than only visually secured. As a result, even if a harmful "Dark Pattern" is being used to trick a user, the system is technically not capable of retrieving their raw license data, proving these manipulative design strategies useless.

2.3.7 Implementation and Security Test of Zero-Knowledge Protocols on SSI Blockchain.

Self-Sovereign Identity (SSI) is presented as a decentralized digital identity model that shifts control from centralized certification authorities to individual users by leveraging blockchain technology together with Zero-Knowledge Proofs (ZKP). In this model, Zero Knowledge protocols allow the tester to demonstrate knowledge of a secret, such as a private key without disclosing it, which limits unnecessary exposure of personal data in distributed environments like cloud services and online platforms. Moya et al. (2023) designed an educational desktop application that implements the Feige Fiat Shamir, Guillou Quisquater, and Schnorr protocols step by step, enabling students to adjust parameters, observe verifier-tester interactions, and analyze how these settings influence both security and efficiency in authentication for public-private key systems. Through

simulated brute-force and man-in-the-middle attacks, their study shows that correctly configured Zero-Knowledge protocols on a blockchain-based SSI infrastructure can provide an auditable yet privacy-preserving approach to digital identity verification (Moya et al., 2023).

2.3.8 The Power and Potential of Zero-Knowledge Proofs.

In the study, (Underwood, 2025) establishes that Zero-Knowledge Proofs (ZKPs) have evolved from theoretical mathematics into a premier cryptographic solution for industrial verification. She notes that these proofs allow digital systems to “demonstrate membership in a language without conveying any additional knowledge,” effectively resolving the inherent conflict between transparency and privacy. Additionally, (Underwood, 2025) finishes that modern identity platforms are increasingly adopting these protocols to provide irrefutable proof of validity, such as verifying a credential, without ever exposing the sensitive personal data underlying the transaction.

2.4 Local Studies

2.4.1 Digital governance in the Philippines: a scoping review of current challenges and opportunities.

A recent study by Andaya highlights that while digital governance in the Philippines has made strides in improving public service accessibility, it remains hindered by critical adoption barriers, specifically in security, infrastructure, and public trust. The study analyzed governance initiatives from 2020 to 2025, as it identifies a significant research gap in the utilization of emerging technologies like blockchain to address these systemic inefficiencies. (Andaya et al. 2025) emphasize that current centralized digital frameworks

have been difficult with “adoption barriers” and “financial capability challenges,” leaving citizens skeptical of the system's integrity. To lessen the risks, (Andaya et al. 2025) recommend that future research prioritizes the development of secure data management architectures that can ensure compliance with privacy standards and transparency.

Synthesis

These studies collectively highlight the critical gap in current digital governance and provide the technical framework for how blockchain can resolve it. They move from identifying the "trust deficit" in the Philippines to demonstrating how Zero-Knowledge cryptography serves as the necessary solution.

To start off, (Andaya et al. 2025) establish the urgency of having blockchain integrated to this research by analyzing digital governance in the Philippines. Their study identifies a significant "future research gap," noting that while accessibility has improved, government systems are still plagued by "adoption barriers" due to a lack of public trust and security. This study sets the stage for the technical solutions explored by (Moya et al. 2023), demonstrating the practical application of Self-Sovereign Identity (SSI) using Zero-Knowledge protocols, where it proved that by allowing users to demonstrate knowledge of a secret without disclosing it, a system can be made "auditable yet privacy-preserving.” Additionally, it is further reinforced by Underwood (2025), who establishes that these Zero-Knowledge Proofs (ZKPs) have evolved from theoretical mathematics into a premier solution for industrial verification, thus effectively resolving the conflict between transparency and privacy.

Furthermore, these studies validate the thesis argument: (Andaya et al. 2025) confirm the importance for a trust-restoring system in the Philippines, while (Moya et al. 2023) and (Underwood 2025) confirm that ZK-SNARKs are the correct algorithm to build it.

METHODOLOGY

This chapter details the methodology employed in the development of the proposed system. It describes the type of research, the project’s architectural design, the hardware and software requirements, the development framework, and the procedures used for system evaluation and data gathering. These sections collectively outline the structured process followed in building and validating the technical prototype.

3.1 Research Type

The study employs a Design and Development Research (DDR) approach supported by applied research and experimental prototyping. DDR is appropriate because the main output of the study is a functional proof-of-concept system demonstrating how zero-knowledge proofs, Sparse Merkle Tree, and hybrid Validium Semaphore mechanisms can be used to achieve transparency and privacy in a government-like licensing workflow. As applied research, the study addresses a real-world problem with closed, non-auditable digital government systems by proposing an open-source architectural solution. As experimental research, the system is developed and tested in a simulated environment to validate the correctness of its verification flows and cryptographic guarantees.

The sections that follow describe the architecture, development environment, algorithms used, system workflow, smart contract components, off-chain operator

logic, testing procedures, and ethical considerations that guided the creation of the prototype.

3.2 System Design

The project follows a design and development approach that produces a functional proof of concept for a privacy preserving licensing compliance workflow that uses zero knowledge proofs with Merkle verification and a hybrid Validium plus Semaphore model. The design simulates a government like setting where a national chain executes smart contracts while a public chain holds the cryptographic anchor, allowing auditors to compare state roots for accountability without exposing personal data. Off chain, a user prepares inputs and generates concise proof, while on chain contracts verify the proof, authenticate signatures, and update a Sparse Merkle tree that records system state for public verification. The model targets the LTO licensing process to show how open verification and privacy can coexist, and it is scoped as an educational and architectural blueprint rather than a production deployment.

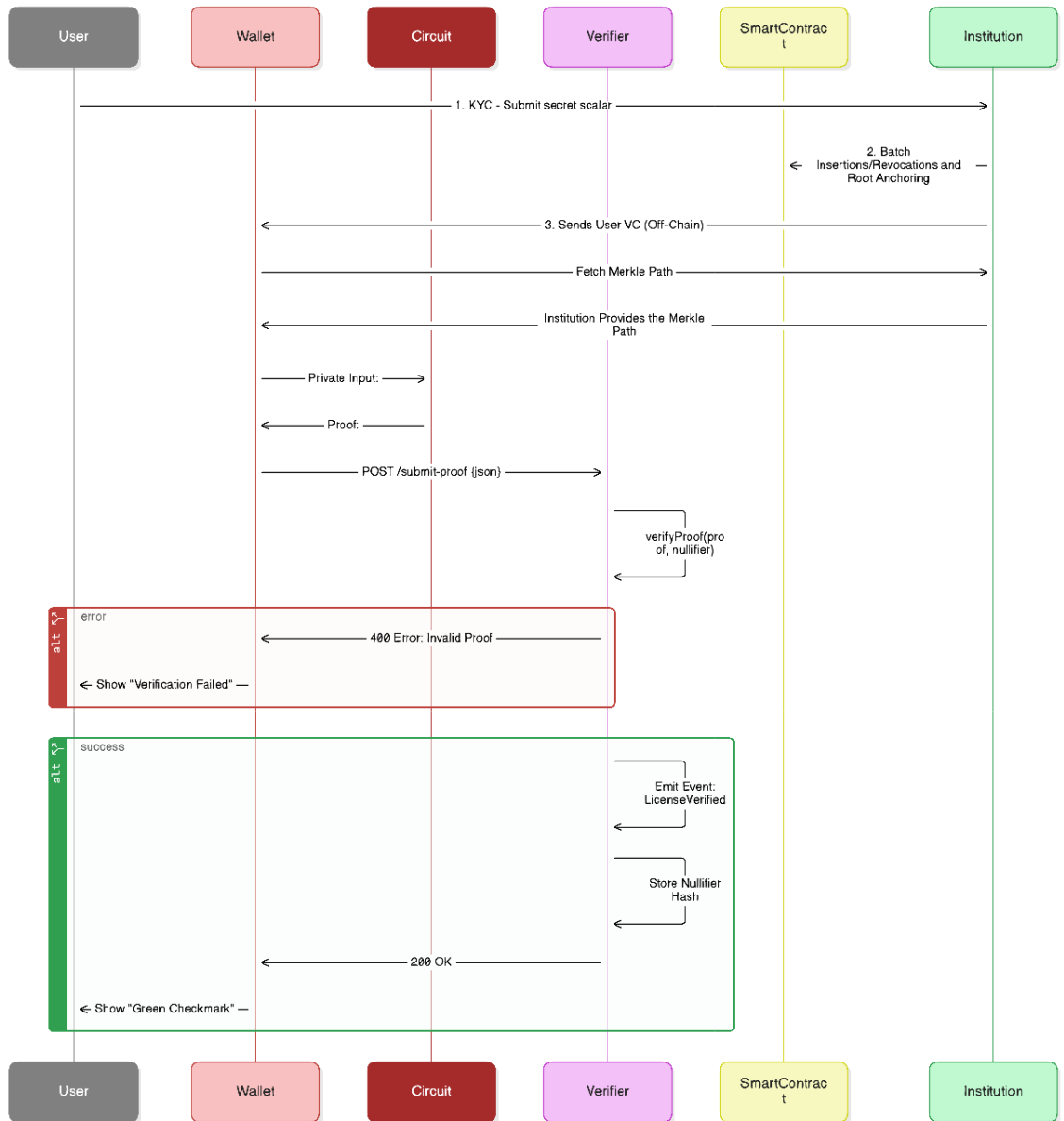


Figure 3.2.1 Sequence Diagram

The Figure 3.2.1 sequence diagram illustrates how the Hybrid Transparency framework works. It details the chronological interactions between the Citizens, Government Administrator, Off-Chain Databases, and the Public Blockchain Ledger. The process initiates whenever a user submits a request or document, which is then first processed within the secure-off-chain environment to ensure strict adherence to data privacy standards. Upon validation by the administrator, the system executes a

cryptographic hashing function to generate a unique digital fingerprint of the record, which is then aggregated into a Merkle Tree structure. Crucially, the diagram visualizes the system's "anchor" mechanism, where only the Merkle Root is transmitted to the immutable public blockchain while the sensitive raw data remains isolated in the private database. The sequence concludes with the public verification phase demonstrating how an external auditor can query the smart contract to perform a Proof of Inclusion, confirming the integrity of the government record without ever accessing the confidential information stored off-chain.

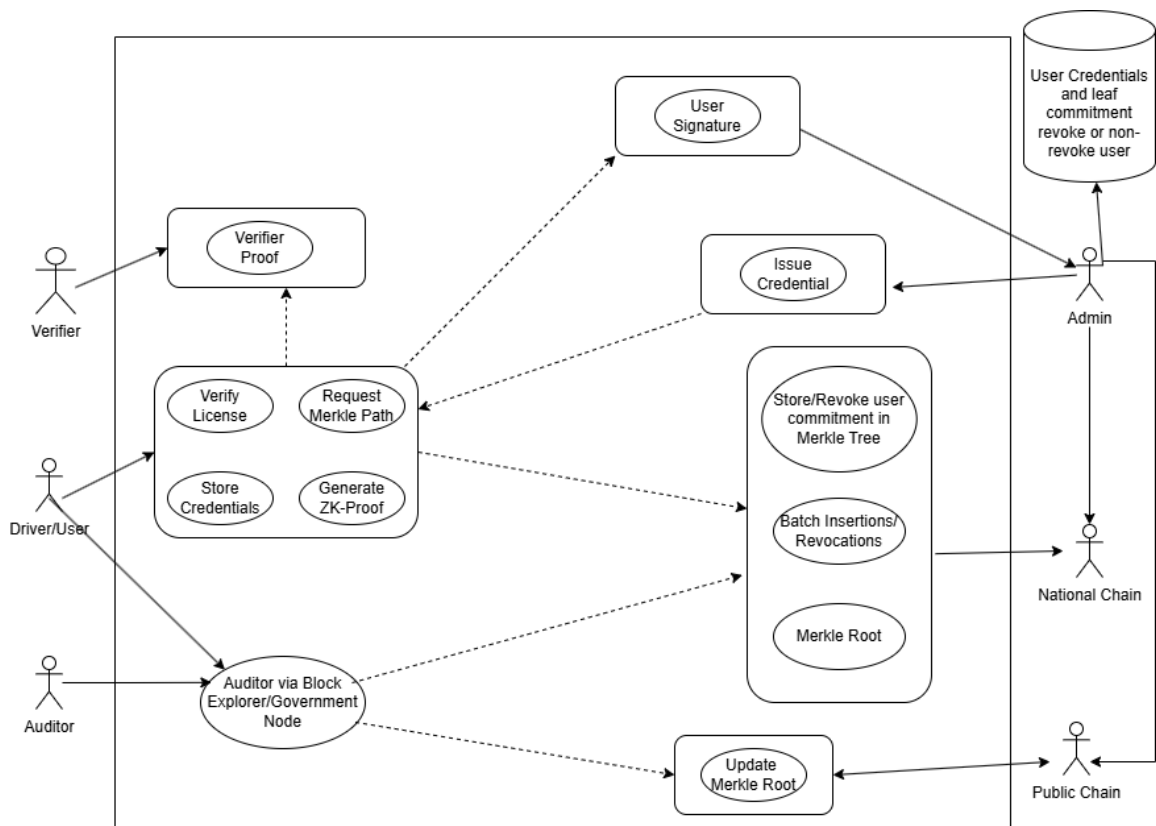


Figure 3.2.2 Use-Case Diagram

The Figure 3.2.2 use case diagram illustrates the distinct functional roles and high-level interactions between primary actors such as the Government Administrator, the Citizen, the External Auditor, and the Hybrid Transparency platform. The use case diagram visualizes the segregation of privileges inherent in the proposed architecture: the Administrator is granted exclusive rights to execute such as “Write” operations or more specifically the functions of uploading digital records and anchoring Merkle Roots to the blockchain. Meanwhile, the Citizen and the Auditor are restricted to “Read” and “Verify” operations, which only allows them to generate Proofs of Inclusion and audit the public ledger without accessing the underlying private database. This visual representation underscores the system’s compliance with the principle of least privilege, ensuring that while the state

maintains control over data management, the verification process remains decentralized and accessible to the public for accountability purposes.

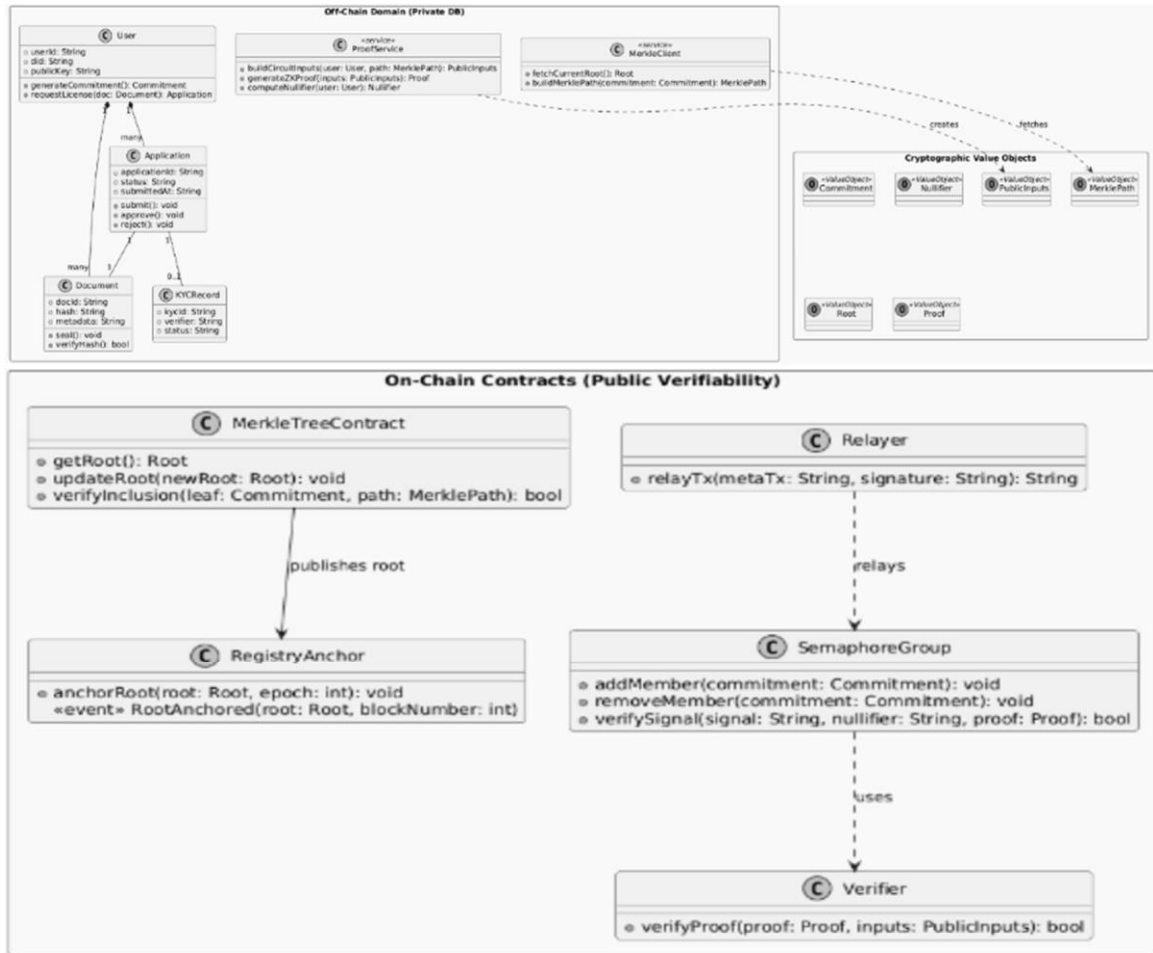


Figure 3.2.3 Class Diagram

The Figure 3.2.3 illustrates the mapping of the static structure of the system by defining the attributes, methods, and relationships of the core software components. The class diagram highlights the architectural split between off-chain and on-chain entities. The primary classes include the Document and User entities, which encapsulate sensitive and private data stored securely off-chain, and the Smart Contract and Merkle Tree classes, which manage the public verification logic on the blockchain. The relationships depicted

emphasize how the Verifier class interfaces with the Proof object to validate transactions without the need of exposing raw attributes. This diagram also demonstrates how the system maintains data integrity through inheritance and encapsulation, ensuring that the cryptographic state on the public ledger accurately reflects the private database without data leakage.

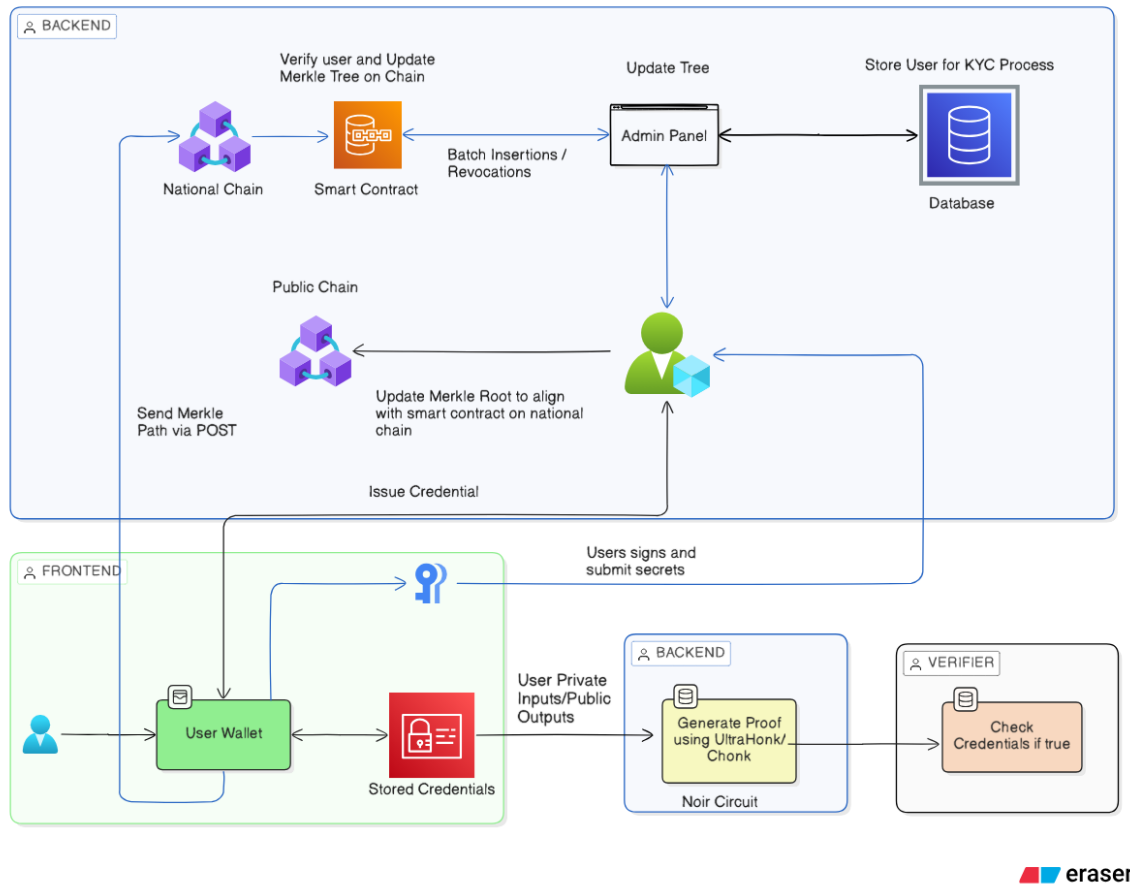


Figure 3.2.4 System Architecture

The Figure 3.2.4 combines Semaphore for anonymous membership verification, ZK-SNARK proofs for correctness and Sparse Merkle tree that commits system state as a single root for efficient inclusion checks. A Validium is used to upload the Merkle root of the public chain to match the national chain root hash. Poseidon hashing and ECDSA signatures to protect data and authenticate actions and publishes an auditable root on a public network so any inconsistent change on the national chain actions detectable. This composition delivers accountability through publicly anchored proofs while preserving citizen confidentiality during the entire process. The system is implemented as a fully mobile-responsive web application, ensuring usability across smartphones, tablets, and desktop devices without requiring native application installation.

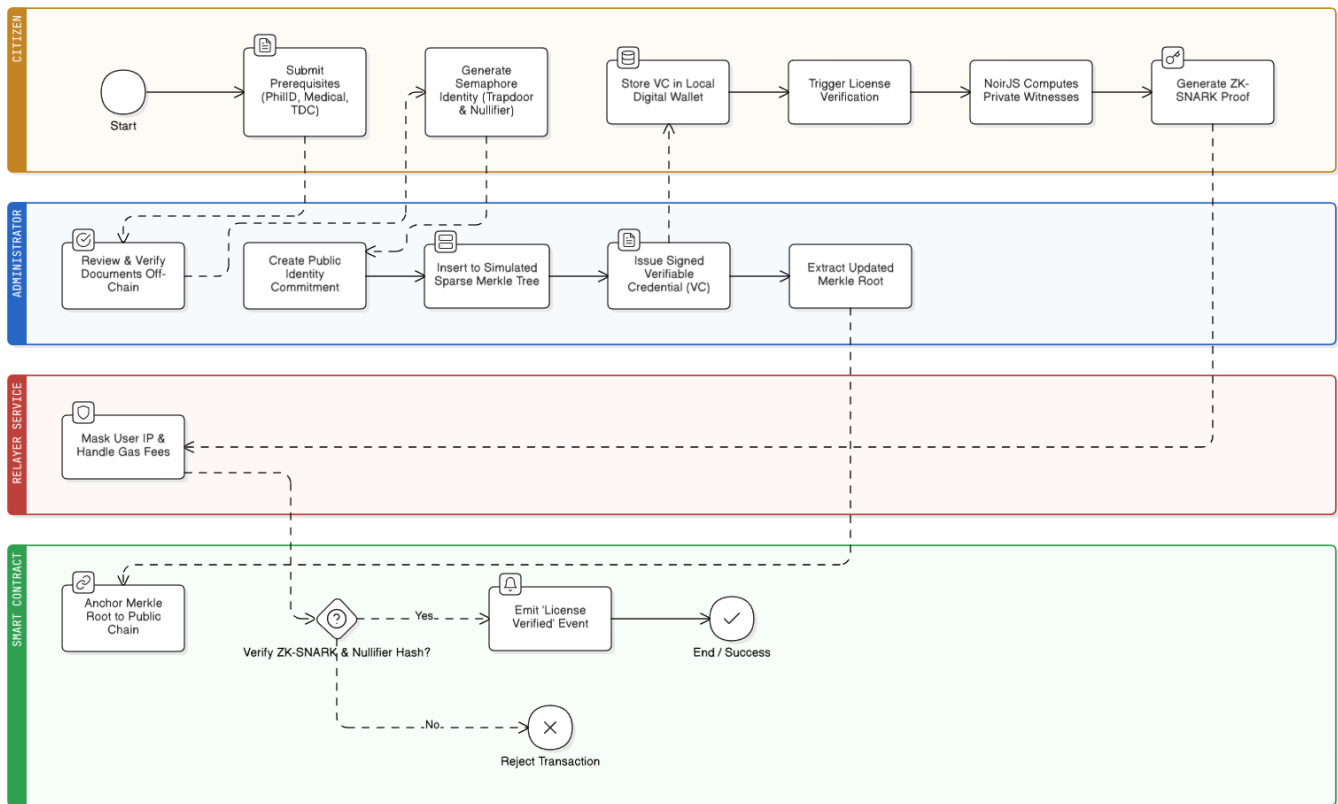


Figure 3.2.5 Zero Knowledge Compliance Pipeline

Figure 3.2.5 depicts the zero-knowledge compliance pipeline adopted in this study. Citizens from the local application can generate cryptographic proofs to securely verify their license status. Each transaction utilizes Verifiable Credentials and zero-knowledge circuits that are to be validated by the proposed smart contract on the Sepolia Testnet.

Phase 1: Off-Chain Prerequisite Ingestion (KYC). Citizens submit prerequisites (PhilID, Medical Certificate, TDC) through the prototype's front-end web application, which simulates a secure government portal. The administrator performs document review and verification through an off-chain process to maintain strict compliance with Data Privacy Act.

Phase 2: Identity & Credential Issuance. The application used by the user creates a Semaphore Identity which consists of a Trapdoor and Nullifier to establish a public Identity Commitment. The system receives this commitment which is added to its simulated Sparse Merkle Tree while a Verifiable Credential (VC) is issued directly to the user's wallet.

Phase 3: Public State Anchoring (Validium). The administrator extracts the updated Merkle Root and publishes it to the Ethereum blockchain. The system achieves mathematical security for the registry's current state through its design, which prevents any upload of unprocessed citizen information.

Phase 4: Client-Side ZK-Proof Generation. The system verifies their valid status because it transfers the processing work to the user's personal device. A NoirJS circuit processes the VC, cryptographic secrets, and the anchored Merkle Root to generate a ZK-SNARK proof, confirming their status without revealing their identity.

3.3 Hardware and Software Requirements

The system utilizes a web-based side for the end user and admin, while simultaneously running the local Blockchain simulation in a Hardhat Environment on a desktop acting as a server or node. The project utilizes the Zero Knowledge Proof algorithm specifically the UltraHonk (ZK-SNARK). Which is system demanding for the user-side, below is the list of requirements to run the algorithm efficiently.

Table 1. Hardware Requirements (for desktop)

Component	Minimum	Recommended
CPU	Quad-core (Intel i5 / Ryzen 3)	6–8 cores (Ryzen 5/7, Intel i7)
RAM	8 GB	16-32 GB
GPU	Optional	NVIDIA RTX 3060/3070/ A-series
Storage	SSD	NVMe SSD

Table 2 Hardware Requirements (for mobile)

Component	Minimum	Recommended
CPU	4-core ARMv8 (Snapdragon 660 / Helio G85 tier)	8-core ARMv9 (Snapdragon 870 / Tensor G2 / Apple A15 tier)
RAM	4 GB	8 GB
GPU	Optional	Required for accelerated proving if supported (Adreno 600-series / Mali-G610)
Storage	Standard SSD / UFS 2.1	UFS 3.1 / UFS 4.0 (faster I/O for proving buffers)

Table 3. Operating System

Operating System	Requirement
Mac OS	Tahoe (26)
Windows	Windows 11

Linux	Ubuntu (20.4+)
Android	Android 10 (API 29)
IOS	IOS 15

Table 4. Browser Requirements

Browser	Requirement
Google Chrome	Latest Version
Firefox	Latest Version
Safari	Latest Version
Microsoft Edge	Latest Version

Table 5. Networking requirements

Component	Recommendation
Internet Connection	A stable connection (Wired/Wireless) that can meet the data consumption requirement of 3 MB per hour

Table 6. Software Requirements

Component	Recommendation
Front-End Software	Nodejs
Back-End Software	Noir Language, Hardhat Developer Environment, MongoDB

3.4 Performance Efficiency Benchmarks

The webapp proving and verify architecture performance efficiency testing uses the Mopro development team's benchmark data which they released in their technical report Mopro x Noir Powering Mobile Zero-Knowledge Proofs. The framework test Zero-Knowledge proofs on devices so developers need to establish proof generation latency baselines which users can test across public and private sector applications.

The benchmark measures the time required to prove and verify a cryptographic operation. The compiled Noir circuit used for this operation contains approximately 12500 constraints. The benchmark measures circuit execution time because its mathematical size stays constant through all testing environments which run the proof generation and verification processes.

Table 4. UltraHonk Performance Benchmark (for Desktop)

JWT Operation	Prove (in seconds)	Verify (in seconds)
Laptop w/o GPU	e.g. 5 secs.	e.g. 5 secs.
Laptop w/ GPU (RTX 3050 ti)	e.g. 5 secs.	e.g. 5 secs.

Table 5. CHONK Performance Benchmark (for Mobile)

JWT Operation	Prove (in seconds)	Verify (in seconds)
POCO Phone	e.g. 5 secs.	e.g. 5 secs.
iPhone 14	e.g. 5 secs.	e.g. 5 secs.

Mopro's benchmarks demonstrate that web applications (WebApps) require approximately 30 seconds to process heavy cryptographic operations. In contrast, their native applications utilize hardware acceleration to execute these same tasks much more efficiently. The Hybrid Transparency prototype currently operates as a WebApp which expects the 30-second benchmark. Mopros demonstrated that moving to a native system would complete proof generation and verification within 5 seconds depending on the hardware. The system protects user data through client-side offline cryptography while native integration will provide users with an experience without interruptions.

3.5 Methods in Developing Software

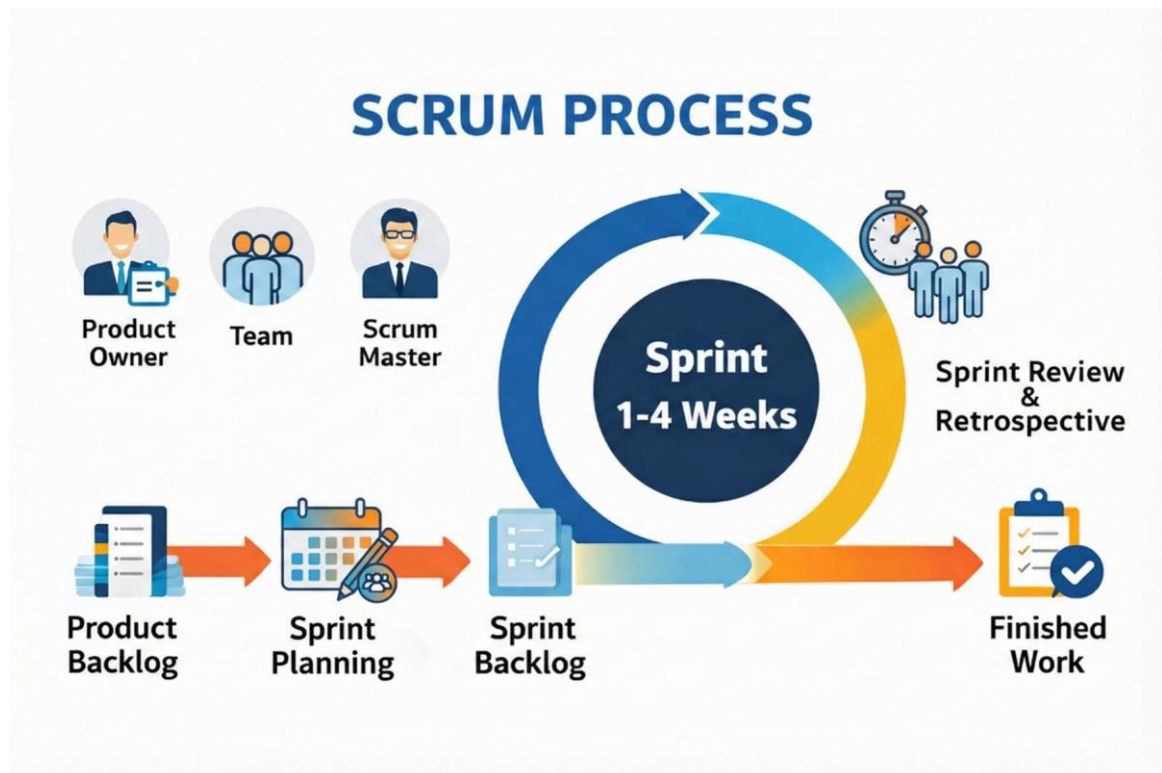


Figure 3.5.1 Scrum/Agile Methodology

The Figure 3.5.1 shows how to create and complete the intended quality of the system with up-to-date information, the authors chose to implement Agile Scrum methodology. It is a project management method that entails constant collaboration and iterative development. It is composed of numerous iterations of the system based on a cycle of Designing, Building, Testing, and Reviewing of the program.

The team is guided and advised by the scrum master who was responsible for reviewing and improving the project according to the time frame. Each week a meeting took place to discuss the action to be taken to develop the project in a gradual manner. Each activity that was planned served as the sprint backlogs that should be accomplished by the team within the week. This process continued to cycle which iterates the system until it reached its complete stable version that will satisfy the product owner.

3.5.2 Algorithm

The World Economic Forum revealed that a full transparent blockchain is not feasible (Lannquist et. Al., 2020), given the nature of the technology to combat corruption, the transparency of the system brought unforeseen consequence to the user side. To further add the difficulty they encountered, implementing an Ethereum

public chain skyrocketed their “Gas Fees.” Which adoption finally became hopeless. Their conclusion was the use of a “hybrid permission-permissionless system”. Stating that both had their strength and weaknesses. Where ZK-SNARK had made its debut on the paper, concluding that Zero Knowledge Proof Algorithms are the solution to the transparency paradox. Though, the paper, from 2020, where Zero Knowledge Proof are still considered at its infancy, was not production ready as the skills for this emerging technology are still critically rare. Additionally, their hybrid “solution” proposal stated L2’s was the solution to rising “Gas Fees” were still considered experimental that time.

The system this project propose is the continuation of the WEF legacy on its fight against corruption. Where the needed technology has finally ripened to solidify its purpose, filling the gap they had lacked. The proposed system operates through a unified privacy preserving verification algorithm that transforms user data into cryptographic commitments, generates a zero-knowledge proof of correctness, and verifies the proof on-chain without revealing sensitive information. The algorithm is composed of multiple cryptographic sub procedures, each described in the following subsections.

$$e: G_1 \times G_2 \rightarrow G_1$$

$$e(g_1^a, g_2^b) = e(g_1, g_2)^{ab}$$

Figure 3.5.2.1 UltraHonk

The formula shown above Figure 3.5.2.1 is the Bilinear Pairing (Boneh et.

al.,2001). Which are the foundations used by UltraHonk. The core cryptographic engine of the system, a high-efficiency Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (ZK-SNARK) protocol. Implemented via the Barretenberg backend library, UltraHonk acts as the orchestration layer that translates the high-level logic of the licensing system into verifiable cryptographic proofs.

Functionally, the system operates through two primary mathematical mechanisms. First, it utilizes an Ultra-Plonk Arithmetization (Aztec Protocol, 2023), which converts the business logic (such as checking if a user exists in the Merkle Tree) into a system of polynomial equations known as an Arithmetic Circuit. Second, to verify these equations without revealing user data, the system employs a Polynomial Commitment Scheme based on Bilinear Pairings. This pairing mechanism allows the smart contract to cryptographically confirm that the arithmetic constraints hold true using elliptic curve operations on the BN254 curve, ensuring both security and privacy.

$$Root = H(H(H(A) + H(\emptyset)) + H(H(\emptyset) + H(D)))$$

Figure 3.5.2.2 Sparse Merkle Tree

The Figure 3.5.2.2 shows a Merkle tree is constructed by recursively hashing pairs of nodes until a single root value is obtained. In a Sparse Merkle tree, new leaves are added over time while preserving the previously committed structure. Figure 2 illustrates a Sparse Merkle Tree, where fixed leaf positions may

either contain a commitment or a predefined null value (Dahlberg, 2016). In the example shown, leaf A contains a valid commitment while leaf B represents a null node; similarly, leaf C is null whereas leaf D contains a commitment. The presence of null nodes enables efficient revocation and membership updates, a property that is essential in privacy preserving identity systems.

For each user, a commitment is inserted at a deterministic position, and successive hashes from the leaf to the root form a unique Merkle path, which serves as proof of inclusion. This Merkle path is supplied as a private witness during ZKSNARK proof generation, while the Merkle root is treated as a public input. Along with the commitment, a nullifier is generated to prevent duplicate applications, ensuring that a user cannot submit multiple valid proofs for the same licensing process. The generated proof is submitted to the on-chain smart contract, which recomputes the Merkle path and verifies that the resulting root matches the stored root. Upon successful verification, the system updates the Merkle tree state, while the applicant proceeds to off-chain KYC and administrative approval.

$$X^0 = (m_1, m_2, \dots, m_r, 0, \dots 0)$$

Figure 3.5.2.3 Poseidon Hash

Figure 3.5.2.3 shows that the Poseidon is a sponge construction over a finite field. It operates on a state vector of width and applies rounds of linear + nonlinear transformations. For the cryptographic commitment scheme, this study utilizes the

Poseidon Hash Function (Grassi et al., 2021). Unlike traditional hash functions such as SHA-256, which rely on bitwise operations that are computationally expensive to prove in Zero-Knowledge circuits, Poseidon is designed specifically for Arithmetic Circuit Efficiency.

It operates natively over the scalar field of the elliptic curve, reducing the constraint complexity by a factor of 100x compared to SHA-256. This optimization is critical for the LTO system to ensure that the client-side proof generation on an end device remains under 2 seconds.

$$s = k^{-l} (H(m) + rd) \bmod n$$

Figure 3.5.2.4 secp256k1 (Ethereum compatibility)

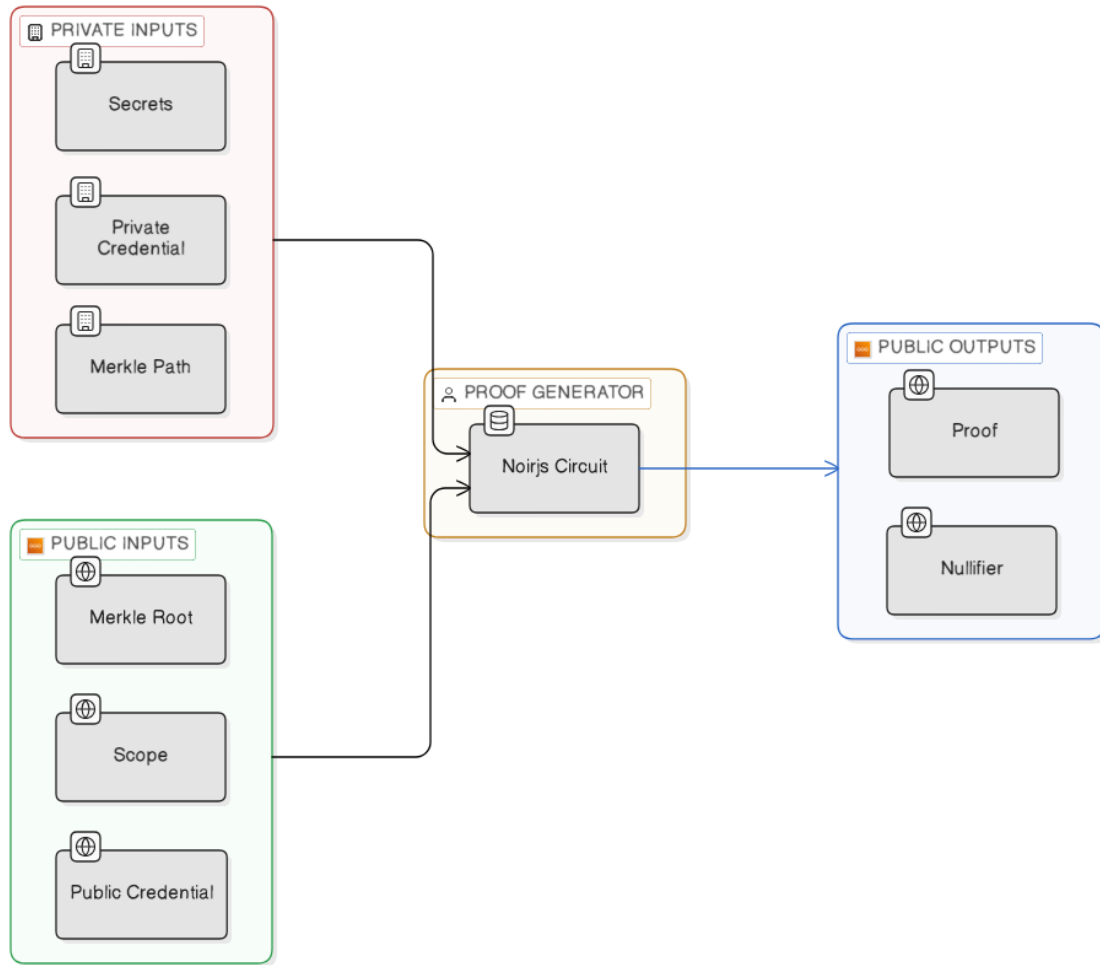
Figure 3.5.2.4 illustrates that to authenticate administrator-initiated state updates and provide non-repudiation within the prototype; the system employs the **Elliptic Curve Digital Signature Algorithm (ECDSA) specifically utilizing the secp256k1 curve**. In practice, each administrator holds a key pair generated over this curve; transactions are signed client-side with the private key, and the verifier validates signatures against the registered public key before accepting any state transition.

This implementation ensures full **Ethereum compatibility**, allowing for seamless integration with EVM-based smart contracts and existing cryptographic

libraries. We adopt the **secp256k1** curve because it is a widely standardized scheme in decentralized systems that offers high efficiency and strong security at 256-bit key sizes. By utilizing this specific curve, only signatures produced by authorized admin keys can affect write operations, establishing an auditable, tamper-evident, and blockchain-ready trail for administrative actions.

3.5.2.5 CHONK (Client-side Highly Optimized plonK)

CHONK is the mobile-optimized proving backend built on top of the UltraHonk ZK-SNARK system. Unlike UltraHonk, which targets desktop-class hardware, CHONK is designed specifically for resource-constrained mobile environments using WebAssembly (WASM). Since CHONK does not introduce new cryptographic primitives or mathematical constructions, it inherits all algebraic structures, polynomial commitments, and pairing-based verification equations defined under UltraHonk.



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3.5.2.6 The Semaphore v4 Protocol: Proof Validation

Figure 3.5.2.6 deepens the system understanding of how signaling establishes a strict cryptographic boundary using the NoirJS domain-specific language (DSL) and the UltraHonk/CHONK proving backend. The circuit is designed to ensure that personally identifiable information (PII) is structurally isolated from the public blockchain by implementing a modern Semaphore v4 signaling protocol. This is achieved by separating the data into Private Witnesses, Public Inputs, and Public Outputs during the client-side proof generation.

1. Private Witnesses

These variables are processed entirely on the user's end device and are strictly constrained, embodying the concepts of the Self-Sovereign Identity framework.

- **Users Secrets:** A high-entropy cryptographic scalar derived deterministically from the user's digital signature. It acts as the foundational root of trust that mathematically binds the user to their credential data without exposing the signature itself.
- **Private Credential inputs:** Specific parameters from the credential (e.g., License Number, Date of Birth) that the user wishes to keep hidden. These are hashed alongside the User Secret inside the circuit to rebuild the identity commitment.
- **Merkle Path:** The array of sibling hashes required to traverse the Sparse Merkle Tree from the user's specific leaf node up to the root. Supplying this path proves the user holds a valid license without revealing their exact index or identity within the database.

2. Public Inputs

These are parameters known to both the prover (citizen) and the verifier (e.g., the traffic enforcer's off-chain device). They provide the context for what is being verified and what data is being explicitly disclosed.

- **Merkle Root:** The current state root of the smart contract anchored on the blockchain. The circuit verifies that the provided Merkle Path mathematically resolves to this exact public root, proving the credential is officially recognized by the issuing authority.
- **Scope:** A specific identifier representing the action being taken. This ensures the generated proof is uniquely bound to a single context and cannot be reused elsewhere.
- **Public Credential:** The unique, cryptographic data bounded to user's credentials and signature. It's reconstructed every time user generates Proof of Inclusion. the cryptographic binding allows that the hash of user's credential is indeed in the Merkle Tree.

3. Noirjs Circuit Logic

Inside the mobile proving environment, the circuit executes three primary constraints:

Rebuilds Leaf: $L = \text{Poseidon}(s, d, N)$

Derives: $\text{Nullifier} = \text{Poseidon}(s, S)$

Verifies: $R = \text{MerklePath}(L, p)$

Where:

- L is the computed Identity Commitment (Leaf).
- R is the computed Merkle Root that is the equivalent to the public State Root.

Data Classification:

- **Private Witness:** The user's secret scalar (s), the hidden credential data (d), and the Merkle sibling path (p).
- **Public Inputs:** The public state Merkle Root (R), the selectively disclosed data (N), and the scope (S), which acts as a public deterministic challenge message.
- **Public Outputs:** The generated *Nullifier*, which the verifier uses to prevent replay attacks for that specific scope.

4. Public Outputs

The circuit executes its constraints and outputs only two mathematically safe variables, which are transmitted as a payload to the Verifier's device.

- **ZK-SNARK Proof:** A succinct, verifiable string of bytes that mathematically guarantees all internal circuit constraints were met without revealing the underlying private witnesses.
- **Nullifier Hash:** An anonymous digital receipt. The verifier's application checks this against the current session; if the same citizen attempts to present the exact

same proof for the exact same checkpoint challenge, the identical Nullifier will be generated, allowing the verifier to automatically reject the replay attack.

$$L = \text{Poseidon}(s, d, N)$$

$$R = \text{MerklePath}(L, p)$$

$$R_{\text{computed}} == R_{\text{public}}$$

$$\text{Nullifier} = \text{Poseidon}(s, S)$$

Where:

- L is the computed Identity Commitment (Leaf).
- R is the computed Merkle Root that is the equivalent to the public State Root.

Data Classification:

- **Private Witness:** The user's secret scalar (s), the hidden credential data (d), and the Merkle sibling path (p).
- **Public Inputs:** The public state Merkle Root (R), the selectively disclosed data (N), and the scope (S), which acts as a public deterministic challenge message.
- **Public Outputs:** The generated *Nullifier*, which the verifier uses to prevent replay attacks for that specific scope.

The implementation of selective disclosure through ZK-SNARKs enhances public accountability by shifting the verification paradigm from "data sharing" to "mathematical proofing." In traditional centralized systems, accountability often requires the bulk collection and warehousing of sensitive Personally Identifiable Information (PII) such as exact birthdates or residential addresses which inadvertently creates massive attack surfaces for data breaches and identity theft. By utilizing the **Hybrid Transparency** model, the system enforces the principle of data minimization; it allows a verifier to confirm a driver's eligibility with absolute mathematical certainty without ever taking custody of the underlying private data. This creates a superior form of accountability where the integrity of the government's state root is publicly auditable on the blockchain, yet the citizen's right to privacy is protected against administrative overreach and systemic leaks. Thus, accountability is achieved not through the exposure of the citizen, but through the immutable transparency of the verification process itself.

It is important to note the architectural lineage of this approach. The Semaphore protocol was originally engineered by the Ethereum Foundation's Privacy and Scaling Explorations (PSE) group to facilitate absolute anonymity in Web3 environments. Its primary real-world use cases include anonymous DAO voting, secure whistleblowing decentralized applications (dApps), and private reputation signaling. However, by adapting Semaphore's robust proof-of-inclusion primitives and integrating a custom flat-hash cryptographic binding, this system successfully repurposes a protocol built

for pure anonymity into a powerful framework for verifiable, selective government identification.

In summary, the Semaphore v4 validation phase represents a fundamental shift in data sharing methodologies for public accountability. Instead of surrendering a physical ID card or a complete digital database record, the user's mobile device transmits a strictly minimized cryptographic payload to the verifier. This transmission consists exclusively of the ZK-SNARK Proof, the derived Nullifier (to prevent replay attacks), and the explicitly authorized Public Inputs (such as the user's Name and the session Scope). The user's secret scalar, hidden credential data, and exact Merkle tree index never leave the local device.

3.6 Methods in Evaluating the System

To ensure interoperability and longevity, this research adopts the **W3C Verifiable Credentials (VC)** and **Decentralized Identifiers (DID)** standards. With these standards, it aligns with current laws and regulations not just locally in the Philippines but also in other parts of the world on data privacy. While the cryptographical process is critically oversee using the three pillars of ZK-SNARKS trust benchmark. This evaluation would not only make blockchain based systems more transparent, but it would also use to compare closed source initiatives and prevent vendor locked in.

3.6.1 Alpha Testing

Alpha testing focused on internal validation of the system as seen in Table 1. represents the user and admins interaction with the system. This includes verifying the correctness of ZK-SNARK proof generation by comparing the inputs if it matches the constraints logics, checking the accuracy of Merkle path computations in a Merkle tree, ensuring proper execution of smart contract verification by deploying in the hardhat environment simulation. And testing basic user interface flows for the administrator and citizen roles.

Table 1. System Behavior

Risk ID	Actor	Category	Test case Description	Expected outcome
----------------	--------------	-----------------	------------------------------	-------------------------

AT-01	User	Proof Generation	User side submits necessary requirements to meet circuit constraints off-chain	Circuit rejects user inputs if constraints are not match.
AT-02	System	Verification	Proof is submitted via POST and verifies through smart contract on-chain	Smart Contract accepts and store user commitments in The Merkle Tree or rejects if user has not met the requirements
AT-03	User/Admin	Revocation Status	User side checks for status (Active or Revoke) and Admin side controls or update status of users.	User unable or able to use Proof of Inclusion depending on status.
AT-04	System	Nullifier	Prevents user from verifying more than once	Smart Contract reject user request if nullifier is already stored on-chain
AT-05	Admin	State Update	Chain update is recorded in the database e.g. Merkle root, Commitments scheme, Revoke users	Databased in sync with the blockchain
AT-06	System	Merkle Path	User paths in the Merkle Tree on-chain	Merkle Path serves as user address in the tree and Proof of Inclusion
AT-07	System	Relayer	Relayer receives proof and forwards to the national chain via POST off-chain	Relayer abstract user identity from chain and relays proofs to the smart contract

As seen in the risks id tallied in Table.1 System Behavior, it encompasses user-system-admin interactions. The list of id risks is used to assess the expected output of the system. Developers would be able to pinpoint the area of concern, separating other components. This ensures that the ZK-SNARKS system must

satisfy three main pillars of trust. Firstly, the COMPLETENESS of the provers private inputs must be able to generate proof if compliance is met. Secondly, SOUNDNESS convinces the verifier that the proof generated by the prover is honest and has not been tampered with. Third, the verifier has ZERO KNOWLEDGE of the prover's private inputs; only the proof and the public inputs are visible to the verifier.

3.6.2 Whitebox Testing

Whitebox testing examines the internal logic of the system. Table 1 represents the key details to assure the system security works as its expected. This includes verifying the correctness of ZK circuit constraints, ensuring Poseidon hashing produced consistent results, confirming the integrity of Merkle path reconstruction within the circuit, and checking the deterministic behavior of the smart contract verifier.

Table 2. Logical Components

Test ID	Component	Internal Logic tested	Input Condition	Expected Behavior
WT-01	Circuit	Merkle Membership Constraint	Input a Leaf that is not part of the provided Merkle Path.	The circuit must fail the assertion root hash
WT-02	Circuit	Nullifier Determinism	Input a Nullifier that is not mathematically derived from the Private Key	The circuit must fail the assertion checking the relationship between the secret and nullifier.
WT-03	Smart Contract	Signal Tampering	Prover attempts to manipulate the Public Signal after witness generation.	The verifier must reject the proof because the public input hash does not match the proof's commitment.

WT-04	Poseidon	Hash Collision Resistance	Input two different preimages to the Poseidon hash function.	The circuit computes distinct outputs confirming no trivial collisions in the finite field.
WT-05	Circuit	Under-Constrained Signals	Audit the circuit for signals that are computed but not constrained (unused witnesses).	All intermediate signals are tied to the public output or a constraint, preventing "Free Variable" attacks.
WT-06	Update Root	Access Control	A wallet address without the ADMIN_ROLE attempts to call the state-changing function.	The contract must revert execution with Access Control: sender is missing role
WT-07	ECDSA	Digital Signature Algorithm	Ensure Credentials and Proofs are signed with Person of Authority	Ensures non-Repudiation by comparing public wallets and credentials Signature

As seen in Table 2. Logical Components, the risks are designed to ensure the internal components do not contradict the architectural design of the cryptographical verification. Ensuring all internal roles are free from errors that would break the system's interactions. Starting from user layer, the inputs must be prepared to accommodate the circuits generation with ECDSA and Poseidon hashing. Along with admin provided Merkle path that ties user into the Merkle tree after verification. In the second layer, the smart contract must be able to prove users' credentials and Proof of Inclusion are tied from other Institutional Merkle Trees. It must also ensure that the users' nullifier is stored after successful verification and rejects Double Spending. And the final layer, which is the admin, must be able to interact with the smart contract logics such as the Merkle tree for reading, deleting

and updating users. Such as Revocations and verifying users off-chain, and issuance of Verifiable Credentials that are tied with admin ECDSA and proof of inclusion to the tree.

3.6.3 ISO-Based Software Evaluation (ISO/IEC 25010)

The system was evaluated following ISO/IEC 25010 quality characteristics. Functionality was assessed through the correct execution of ZK-SNARK proof generation, Merkle path verification, and root anchoring. Reliability was examined by repeatedly submitting proofs to ensure deterministic and fault-free validation. Performance efficiency was measured through client-side Poseidon hashing and proof generation, which remained within acceptable latency bounds. Security evaluation focused on the correctness of cryptographic constraints, resistance to tampering, and protection against unauthorized access. Usability and compatibility were verified through the responsiveness of the web interface and the successful deployment across the Hardhat local chain, Node.js frontend, and MongoDB backend.

Functional suitability - This evaluates whether the core zero-knowledge verification flows execute correctly and completely, ensuring the system processes valid inputs exactly as designed.

Performance efficiency - This measures the speed and resource consumption of the prototype, specifically tracking the latency of off-chain proof generation and the computational cost of on-chain verification.

Reliability - This tests the determinism of the verification process. It involves running basic fault cases and checking error handling to guarantee the system produces consistent, predictable results.

Security - This uses targeted negative tests to validate the zero-knowledge privacy claims. It mathematically guarantees that no personally identifiable information (PII) is ever leaked or stored on-chain.

Usability - This assesses the practical user experience, specifically looking at how easily and efficiently administrators and verifiers can complete their required operational tasks.

Compatibility - This checks the interoperability of the system at the artifact and format boundaries, such as its ability to properly handle standard APIs and output Verifiable Credentials (VC) in JSON format.

3.6.4. STRIDE THREAT MODELING

Table 3.

STRIDE Category	Threat Context	Architectural Mitigation
-----------------	----------------	--------------------------

Spooftng	An attacker attempts to impersonate a legitimate citizen to generate a valid proof of license.	Prevented by the Semaphore Identity . Proof generation requires the user's private Trapdoor and Nullifier secrets, which are generated and stored strictly on the user's local device and are never shared.
Tampering	A malicious actor or corrupt official attempts to secretly alter or delete license records in the private database.	Mitigated by the Validium Architecture . Any unauthorized change to the off-chain data alters the Sparse Merkle Tree's state root. This mismatch will be instantly flagged by the public Ethereum chain where the true root is anchored.
Repudiation	An administrator denies executing a state change, such as approving or revoking a citizen's credential.	Addressed through ECDSA Signatures . All state-changing operations pushed to the Smart Contract (like updating the Merkle Root) require the administrator to sign the transaction, creating an undeniable cryptographic audit trail.
Information Disclosure	The public ledger or a network interceptor exposes a citizen's Personally Identifiable Information (PII).	Prevented by ZK-SNARKs (NoirJS) . The mathematical circuit ensures only a cryptographic proof and a Nullifier Hash are broadcasted to the network. Raw data (like names or exact medical statuses) never leaves the client device.

Denial of Service (DoS)	A spammer attempts to exhaust system resources or flood the network with duplicate verification requests.	Mitigated by Nullifier Hashes and the Relayer . The Smart Contract instantly rejects any transaction containing a previously used Nullifier, preventing "double-signaling." The off-chain Relayer also acts as a buffer to rate-limit incoming requests.
Elevation of Privilege	A standard user or an unauthorized node attempts to insert themselves into the Merkle Tree to grant themselves a license.	Prevented by Smart Contract Access Control. The system strictly enforces Role-Based Access Control (RBAC), ensuring that only wallets explicitly granted the ADMIN_ROLE can execute write functions to update the credential registry.

Table 3. illustrate the vulnerable risk this project may Forsee and how it's alleviated. This study applies the STRIDE threat modeling framework to evaluate security risks at the architectural level of the proposed Hybrid Transparency system. STRIDE was selected because it is a design-time methodology suited for identifying systemic threats in distributed and cryptographic architectures, rather than implementation-level vulnerabilities. The analysis focuses on threats relevant to identity spoofing, unauthorized state manipulation, privacy leakage, and abuse of verification mechanisms, which are critical concerns in government-scale credential systems. Each identified threat is mitigated through built-in cryptographic guarantees such as zero-knowledge proofs, nullifiers, Merkle root

anchoring, and role-based access control. Implementation-level attacks (e.g., side-channel attacks, browser exploits, or network-layer denial-of-service) are considered outside the scope of this prototype and are acknowledged as future work for production deployment.

3.6.5 Data Gathering and Sampling Technique

The sampling technique utilized in this study is purposive sampling. This method was chosen to select respondents who possess the specific technical background and operational understanding required to properly evaluate a decentralized zero-knowledge system. The target evaluators are categorized into two groups: IT/Blockchain Professionals and System Administrators.

Blockchain Professionals

Because the system's architecture is built on ZK-SNARKs, the Semaphore protocol, and the Validium model, individuals with a background in software development, cybersecurity, or blockchain technology were selected. Their primary role is to evaluate the technical robustness of the prototype. Their insights are heavily focused on assessing the system's Security, Performance Efficiency, Reliability, and Compatibility, ensuring the architectural blueprint mathematically protects data as intended.

System Administrators and Verifiers

Since the prototype simulates a government credentialing workflow (such as the LTO licensing process), individuals familiar with system administration, data management, or verification processes were chosen. Their evaluation focuses heavily on the Usability and Functional Suitability of the system. Their feedback helps determine if the platform allows authorities to easily issue and verify credentials without exposing or mishandling sensitive personal data.

3.6.6 Questions for the Survey

There were two main sets of surveys: TAM-based for the public sector respondents (two versions) and ISO-25010-based for the IT and blockchain experts. Both surveys encompass all the sections published in the respective frameworks they are based on.

TAM-Based Survey

Perceived Usefulness

1. The proposed zero-knowledge architecture offers a significant upgrade to current government verification workflows.

2. This hybrid system is flexible enough to handle various official records while strictly maintaining data privacy.
3. Utilizing this framework would allow public sector staff to validate credentials more efficiently without compromising security.

Perceived Ease of Use

1. Operating the verification dashboard is intuitive and does not demand deep technical expertise in blockchain.
2. The step-by-step process of approving or checking a cryptographic proof is logical and easy to follow.
3. Interacting with the prototype to process a simulated transaction was a seamless experience.
4. Verifying credentials with the help from the model's implementation would be easy to accomplish (for system admin only)

Attitude Towards Use

1. Integrating a privacy-first blockchain approach is a highly beneficial step for government data management.
2. The robust security and privacy benefits provided by this architecture easily justify the effort required to implement it
3. Deploying this system would significantly boost citizens' confidence in how their data is handled.

4. I am convinced that this hybrid transparency model holds immense value for the future of e-governance.

Behavioral Intention to Use

1. I am willing to utilize this dashboard for routine credential checks in my daily administrative tasks (for system admin only).
2. I would encourage other Blockchain experts to explore this zero-knowledge framework.
3. I would strongly support integrating this technology into broader national identity initiatives.
4. I would apply the privacy standards demonstrated by this prototype as a baseline for processing sensitive citizen information (for system admin only)

ISO-25010 Based Survey

Functional Suitability

1. The core zero-knowledge circuits successfully compute and validate legitimate inputs without any logical flaws.
2. The mathematical nullifiers effectively trigger deterministic rejections whenever a duplicate verification attempt occurs.
3. The Validium-based design flawlessly isolates private identity markers from the transparent on-chain records.

Performance Efficiency

1. The local simulation handles the cryptographic workload smoothly without causing system failures or resource exhaustion.
2. The turnaround time for compiling an off-chain ZK-SNARK proof falls within an acceptable window for real-world application.
3. The gas consumption and processing power required to anchor the proof to the blockchain are highly optimized.

Reliability

1. The prototype demonstrates strict determinism, yielding identical verification states given the exact same set of parameters.
2. The system architecture remains stable and responsive even after executing multiple consecutive proof generation cycles.
3. The application accurately captures and reports logical exceptions, such as unauthorized access or credential reuse, without crashing.

Security

1. The cryptographic mechanisms guarantee absolute zero leakage of personally identifiable information (PII) to the public ledger.
2. The integration of Semaphore successfully establishes group membership and authorization while maintaining complete user anonymity.
3. The architecture properly isolates the local "Trapdoor" and secret hashes, preventing them from being broadcasted during the verification phase.

Usability

1. The administrative interface abstracts the complex blockchain operations into a highly manageable and intuitive workflow.
2. The learning curve for an official to adopt and operate this zero-knowledge dashboard is minimal.
3. The heavy mathematical processes running in the background do not disrupt or complicate the standard duties of the verifier.

Compatibility

1. The prototype successfully formats the finalized cryptographic proofs into standardized structures like JSON Verifiable Credentials.
2. The Relay Service exhibits flawless interoperability when passing data between the localized off-chain environment and the smart contracts.
3. The system is built with flexible boundaries, demonstrating strong capability to connect with existing external APIs.

3.6.7 Statistical Treatment of Data

To keep the analysis accurate, the encoded survey scores need to be reviewed first to clear out any errors. After verifying the data, the weighted mean gets calculated for each question depending on the respondents. This specific step is for maintaining consistency throughout the research. Using the weighted mean helps us capture a realistic picture of the

central tendency, since it directly measures how often an answer was chosen along with its scale.

The formula for weighted mean:

$$\text{weighted mean} = \frac{\sum wx}{\sum x}$$

Where:

w = the weight or frequency for each data point

x = the value of each data point

Data Privacy Act of 2012

The system complies with RA 10173 by ensuring that all personal data remains exclusively in the off-chain private database. The public chain stores only cryptographic commitments and Merkle roots, preventing disclosure of personally identifiable information. Zero-Knowledge proofs provide selective disclosure, enabling verification without revealing sensitive attributes, consistent with data minimization and proportionality requirements.

RA 8792 – Electronic Commerce Act (Non-Repudiation)

Administrator actions—including document anchoring and state updates—are authenticated using ECDSA signatures. The immutable Merkle root histories recorded on the blockchain provide a tamper evident audit trail, ensuring nonrepudiation of electronic transactions.

W3C Verifiable Credentials Data Model 2.0

The system follows the issuer–holder–verifier model defined by VC 2.0. Cryptographic commitments and Merkle inclusion proofs map to the VC requirement for secure, tamper evident, independently verifiable credentials. The public verifier smart contract serves as an interoperable verification endpoint. Trust Model,

ISO 180135 (Mobile Driving License Selective Disclosure)

ZK-SNARK proof generation enables selective disclosure of user claims, allowing individuals to prove eligibility or membership without revealing raw identity data. This aligns with the ISO mDL privacy and attribute-based verification requirements.

International Benchmarks (OpenCerts, EU eIDAS 2.0)

OpenCerts demonstrates the feasibility of blockchain anchored verification without exposing source records, supporting the Merkle root anchoring design. Likewise, eIDAS 2.0's formal adoption of Zero-Knowledge proofs validates the system's architectural commitment to privacy preserving public verification.

CADENA Act. (Citizen Access & Disclosure of Expenditures for National Accountability)

While the CADENA Act is pending enactment, the Senate approved version mandates a digital budget platform with tamper resistant public disclosure of budget documents and contemplates blockchain secured transparency. The proposed system's public anchoring of Merkle roots and auditable on-chain state aligns with CADENA's transparency and immutability objectives by enabling third party verification of government records without exposing personal data.

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